



RS6130

Hardware Design and PCB Layout

Guidelines

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1 Feature

- **FMCW Millimeter-Wave Radar**

- 58-64 GHz coverage range, 6 GHz scanning bandwidth
- 1 transmitting channel and 3 receiving channels, equipped with AiP packaged antennas
- Integrated RFPLL, DPLL, transmitter, receiver, analog-to-digital converter, and baseband
- HWA is used for 1D/2D FFT, static clutter removal, and CFAR operations

- **2.4 GHz Wireless Connection**

- Protocol: Supports BLE 5.3, BLE Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and 2.4 GHz proprietary protocols
- Bluetooth PHY: 1 Mbps, 2 Mbps, Long-Range S2 (500 Kbps), S8 (125 Kbps)

- **Application System**

- 32-bit RISC-V CPU, featuring FPU (up to 128 MHz) and I/D cache, for radar post-processing, wireless, and applications

- **Host and Device Interfaces**

- x3 UART

- x2 QSPI with XIP

- x2 I2C

- x8 PWM

- x10 GPADC

- **Clock Source**

- 32/40 MHz crystal oscillator as clock reference
- Internal LSI and MSI RC oscillators
- Optional 32 kHz crystal for low-power operation

- **Power Management**

- Single power supply ranging from 1.7V to 5.5 V
- Built-in LDO network for enhancing PSRR
- BOM optimization and power consumption optimization modes

- **Ultra-Low Power Consumption Characteristic**

- Power status: Off, Standby, Idle, and Active (sensing, processing, and connecting)

- **Encapsulation**

- AiP FCCSP: 8.8 x 5.8 mm² with 56 pins

- **Temperature Range**

- Operating junction temperature: -40 °C to 105 °C

2 Pin Description

2.1 Chip Pin Diagram

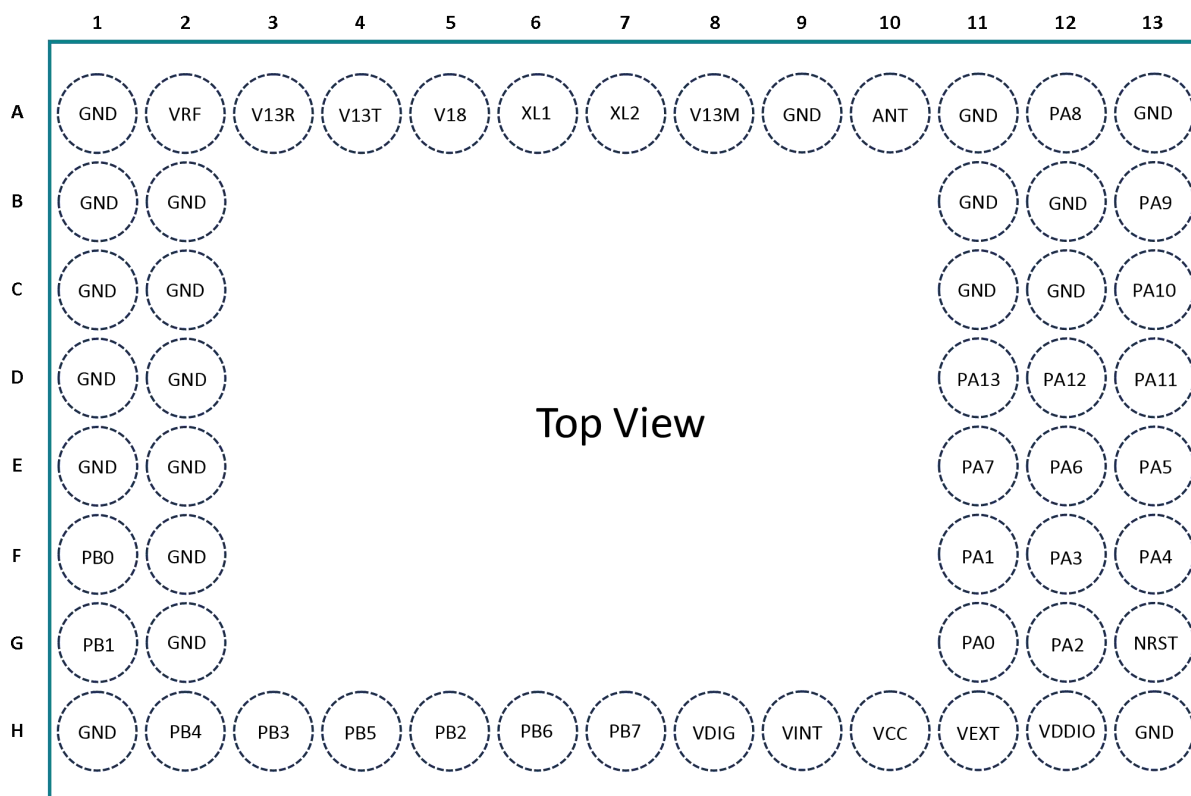


Figure 2-1 Pin diagram

2.2 Pin Description

Table 2-1 Pin Descriptions

Pin name	I/O	Pin number	Remark
VCC	Power In	H10	Power input, 1.7~5.5V
VINT	Power Out	H9	INT LDO output
VEXT	Power Out	H11	EXT LDO output
VDIG	Power Out	H8	DIG LDO output
VDDIO	Power In	H12	GPIO power input, 1.7~3.6V
V18	Power In	A5	MMW analog power supply input, 1.5~2.5V
V13T	Power In	A4	MMW RF transmitting power supply input 1.2~2.5V
V13R	Power In	A3	MMW RF receiving power supply input, 1.2~2.5V
VRF	Power Out	A2	MMW RF LDO output, default 1.0V
V13M	Power In	A8	2.4G analog power input, 1.2~2.5V

XL1	Analog	A6	32/40M crystal oscillator output
XL2	Analog	A7	32/40M crystal oscillator input
ANT	RF	A10	2.4G antenna port
GND	Ground	A1,B1,C1,D1,E1, H1,B2,C2,D2,E2, F2,G2,A9,A11,A 13,B11,B12,C11, C12,H13	Ground Pin, Short directly to module ground plane
NRST	In	G13	Reset the chip, and pull it up to VDDIO by default
PA0	In/Out	G11	GPIO input/output, Wakeup IO
PA1	In/Out	F11	GPIO input/output, Wakeup IO
PA2	In/Out	G12	GPIO input/output, Wakeup IO
PA3	In/Out	F12	GPIO input/output, Wakeup IO
PA4	In/Out	F13	GPIO input/output, Wakeup IO
PA5	In/Out	E13	GPIO input/output, Wakeup IO
PA6	In/Out	E12	GPIO input/output, Wakeup IO
PA7	In/Out	E11	GPIO input/output, Wakeup IO
PA8	In/Out	A12	GPIO input/output, Wakeup IO
PA9	In/Out	B13	GPIO input/output, Wakeup IO
PA10	In/Out	C13	GPIO input/output, Wakeup IO
PA11	In/Out	D13	GPIO input/output, Wakeup IO
PA12	In/Out	D12	GPIO input/output, Wakeup IO
PA13	In/Out	D11	GPIO input/output, Wakeup IO
PB0	In/Out	F1	GPIO input/output, Wakeup IO
PB1	In/Out	G1	GPIO input/output, Wakeup IO
PB2	In/Out	H5	GPIO input and output
PB3	In/Out	H3	GPIO input and output
PB4	In/Out	H2	GPIO input and output
PB5	In/Out	H4	GPIO input/output
PB6	In/Out	H6	GPIO input and output
PB7	In/Out	H7	GPIO input and output

2.3 GPIO Multiplexing Table

Table 2-2 GPIO Multiplexing Table

GPIO Name	Func2	Func3	Func4	Func5	Func6	Func7	Func after rst
PA0/WIO0	SPI0_CS	UART1_TX			PWM0		OFF/No-Pull
PA1/WIO1	SPI0_CLK	UART1_RX			PWM1		OFF/No-Pull
PA2/WIO2	SPI0_MOSI	UART1_RTS	I2C0_SDA	UART2_TX	PWM2		OFF/No-Pull
PA3/WIO3	SPI0_MISO	UART1_CTS	I2C0_SCL	UART2_RX	PWM3		OFF/No-Pull
PA4/WIO4	SPI0_HOLD	I2C0_SDA			PWM4		OFF/No-Pull
PA5/WIO5	SPI0_WP	I2C0_SCL			PWM5		OFF/No-Pull
PA6/WIO6	UART1_RTS	UART2_TX	I2C1_SDA		PWM6	GPADC6	OFF/No-Pull
PA7/WIO7	UART1_CTS	UART2_RX	I2C1_SCL		PWM7	GPADC7	OFF/No-Pull
PA8/WIO8/LX L1 ¹	UART1_TX	I2C1_SCL			PWM0	GPADC0	OFF/No-Pull
PA9/WIO9/LX L2 ¹	UART1_RX	I2C1_SDA			PWM1	GPADC1	OFF/No-Pull
PA10/WIO10	I2C1_SDA				PWM2	GPADC2	OFF/No-Pull
PA11/WIO11	I2C1_SCL				PWM3	GPADC3	OFF/No-Pull
PA12/WIO12	UART2_RX	UART1_RTS			PWM4	GPADC4	OFF/No-Pull
PA13/WIO13/TEST ²	UART2_TX	MCO_CLK			PWM5	GPADC5	IN/UP
PB0/WIO16	UART0_TX ³				PWM0		OFF/UP
PB1/WIO17	UART0_RX ³				PWM1		OFF/UP
PB2	UART0_RTS	FC_WP	SPI0_WP		PWM2		OFF/UP
PB3	UART0_CTS	FC_HOLD	SPI0_HOLD		PWM3		OFF/UP
PB4	FC_CS	UART2_RX	SPI0_CS		PWM4		OFF/No-Pull
PB5	FC_CLK	UART2_TX	SPI0_CLK		PWM5		OFF/No-Pull
PB6	FC_MOSI	I2C0_SDA	SPI0_MOSI		PWM6		OFF/No-Pull
PB7	FC_MISO	I2C0_SCL	SPI0_MISO		PWM7		OFF/No-Pull

1. LXL1 and LXL2 are 32K crystal oscillator pins that are multiplexed with PA8 and PA9. When used for analog functions, the GPIO function needs to be disabled.
2. When the chip is powered on for the first time or restarted after resetting, PA13 will be configured in input and pull-up mode. If this IO is pulled to GND at this time, it will enter ATE test mode. Therefore, PA13 should be avoided being pulled low.
3. The chip uses UART0 of PB0 and PB1 as the default download and control serial port by default.

3 Peripheral Interface Characteristics

3.1 GPIO

The chip provides 22 GPIO, all of which have complex multiplexing relationships.

The GPIO features are as follows:

- Input, output, and high impedance modes
- The input mode supports pull-up and pull-down, while the output mode supports push-pull and open-drain
- The IO drive capability can be configured to 20mA, 40mA, 60mA, and 80mA
- All IO supports interrupt mode
- 16 IO (PA0~PA13, PB0~PB1) support low-power sleep maintenance and interrupt wake-up functions

3.2 UART

The chip provides three UART controllers: UART0, UART1, and UART2. UART0 and UART1 have hardware flow control capabilities.

The UART features are as follows:

- Compatible with industrial standard 16550
- Support hardware flow control mode
- Support baud rate self-adaptation
- Support DMA
- Supports fractional baud rate, with a maximum baud rate of 3Mbps
- Support low-power sleep wake-up

3.3 IIC

The chip provides two IIC controllers, IIC0 and IIC1.

The characteristics of IIC are as follows:

- Supports both master and slave modes
- Supports standard mode at 100Kb/s, fast mode at 400Kb/s, and enhanced fast mode at 1Mb/s
- Supports both 7-digit and 10-digit addresses
- Support DMA
- Support low-power sleep wake-up

3.4 SPI

The chip provides a QSPI controller that supports full-duplex, half-duplex, and serial simplex transmission.

The characteristics of QSPI are as follows:

- Supports both master and slave modes
- Maximum operating frequency: 96MHz
- Supports single-line, dual-line, and quad-line
- Configurable clock polarity and phase
- Support DMA

- Support host mode delayed sampling
- Support low-power sleep wake-up

3.5 PWM

The chip provides four PWM generators, each of which outputs two PWM signals and a pair of complementary signals.

The PWM characteristics are as follows:

- 8-channel PWM output and 4-channel capture input (PWM0, PWM2, PWM4, and PWM6)
- Configurable dead zone
- Supports pulse signal, periodic signal, and complementary signal output
- Adjustable duty cycle from 0-100%

3.6 GPADC

The chip provides a 12-bit resolution and a 10-bit effective bit SAR ADC converter.

The characteristics of GPADC are as follows:

- 8-channel multiplexed ADC
- Support calibration and compensation
- The reference voltage is 1.2V, supporting voltage attenuation in 2/3, 1/2, 1/3, 1/5, and 1/9 settings, with a maximum input voltage of 3.3V
- Oversampling configurable: 16, 32, 64, 128
- Support data comparison and data interruption
- Supports four working modes: single mode, single cycle mode, continuous mode, and burst mode

4 Peripheral Interface Usage Guide

4.1 GPIO Usage Notes

Some GPIO have special characteristics, and the following items should be noted in application schemes:

- 1) PA0~PA13 and PB0~PB1 have WIO (Wire Input/Output) function, supporting system wake-up through interrupt in low-power scenarios, while other GPIO do not have this function;
- 2) PA13 has the function of entering the chip ATE test mode (this function is only used for chip production testing and is not applicable to solution development). When the chip NRST is pulled low or powered on for the first time, PA13 is in a weak pull-up state. When NRST is pulled high to release or when the chip is powered on for the first time, PA13 cannot be pulled low. After the chip starts up, PA13 can be used normally;
- 3) When the chip is powered on or reset and starts up, PB0 and PB2 output high voltage levels, maintaining this state for tens of microseconds;
- 4) After the chip is powered on or reset and starts up, PA0 and PA1 are in a weak pull-up input state.
- 5) For unused IO, leave them unconnected.

4.2 Recommended Interface Usage

4.2.1 Recommended UART Interface Usage

- 1) PB0 and PB1 are the default UART0, serving as the chip programming firmware interface and communication interface, and supporting low-power wake-up function;
- 2) PB2~PB7 are dedicated external Flash interfaces for applications;
- 3) PA0 is the firmware's default UART1-TX reporting data interface. In application design, PA0 is preferably considered as a debug or reporting data interface.
- 4) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.
- 5) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

4.2.2 Recommended IIC Interface Usage

- 1) PA4 and PA5 are the default IIC0, serving as the chip programming firmware interface and communication interface, and supporting low-power wake-up function;
- 2) APB0 is the trigger signal for the chip to enter forced programming mode.
- 3) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.
- 4) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

4.2.3 Recommended SPI Interface Usage

- 1) PA0~PA3 are the default SPI0, serving as the chip programming firmware interface and communication interface, and supporting low-power wake-up function;
- 2) PB2~PB7 are dedicated external Flash interfaces for applications;
- 3) PB0 is the trigger signal for the chip to enter the forced programming mode.

- 4) The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.
- 5) To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.
- 6) Note: If the SPI operating frequency is above 30MHz, please ensure that the SPI controller on the master side is equipped with data sampling delay adjustment capability to ensure correct reading of radar data at different frequencies.

4.3 Enter Programming Mode

Based on the specific hardware configuration, users can enable the chip to enter the programming mode through the following two methods.

Option 1: The main controller can control the NRST pin through IO

- 1) Keep VCC powered on, with PB0 and PB2 at high voltage level;
- 2) Set NRST to low level and delay for 20ms;
- 3) Set NRST to high level and delay for 50ms;
- 4) Set PB0 (or PB2) to low level, and set NRST to low level;
- 5) Delay for 20ms, set NRST to high level;
- 6) Delay for 50ms, set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered programming mode.

Option 2: Suitable for systems with an external control power supply interruption mechanism

- 1) Power off our chips, Flash, or modules. If the external capacitor is 10uF, it is recommended to maintain the power-off state for 10 seconds. Please adjust the power-off duration based on the external capacitor and the actual power-down speed;
- 2) Set PB0 (or PB2) to low level;
- 3) Power on our chip and Flash or module again, delay for 50ms, and set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered programming mode.

4.4 Programming Interface

The chip provides three programming interfaces for firmware, namely UART, IIC, and SPI, and adopts a polling method to configure the communication interface. The default programming interface configuration is as follows:

- 1) UART uses the UART0 interface of PB0 and PB1
- 2) IIC uses the IIC0 interface of PA4 and PA5
- 3) SPI uses the serial simplex mode of PA0~PA3
- 4) After the chip enters the forced programming mode, the UART, IIC, and SPI interfaces perform polling and handshake communication. Once the communication is successful, the firmware is downloaded and programmed.

5 Power-on Sequence

5.1 Power-on Sequence

This chip has no special requirements for the power-on sequence of various power supplies, but there are still the following considerations for different configuration schemes:

1) If SPI NOR Flash is used as the boot device, and the NRST signal is pulled up to VDDIO by default, it is not controlled by external signals:

After the chip is powered on and NRST is released (voltage climbs to high level), it will immediately start and attempt to read the firmware in the SPI NOR Flash. On the other hand, the SPI NOR Flash needs to undergo a specified duration of power-up before it can be accessed. Generally, this chip solution uses VDDIO to power the SPI NOR Flash. Therefore, an RC delay circuit needs to be added around the NRST pin (refer to **Figure 7-1 Reference schematic diagram**) to ensure that during the power-up process, the SPI NOR Flash is powered up and undergoes the required delay before the chip's NRST signal is released.

Please ask the user to calculate the parameters of the RC delay circuit according to the power-on timing requirements specified in the SPI NOR Flash specification.

2) If SPI NOR Flash is used as the boot device and NRST is controlled by an external master, the chip can be reset and booted by manipulating NRST after power-on is completed and the external master is ready, ensuring that SPI NOR Flash has completed power-on.

6 Package Description

6.1 Package Block Diagram

RS6130 adopts FCCSP packaging, with a bump diameter of 350um, pitch of 650um, standoff of 280um, dimensions of 8.8 x 5.8 x 0.68 mm³, and 56 balls.

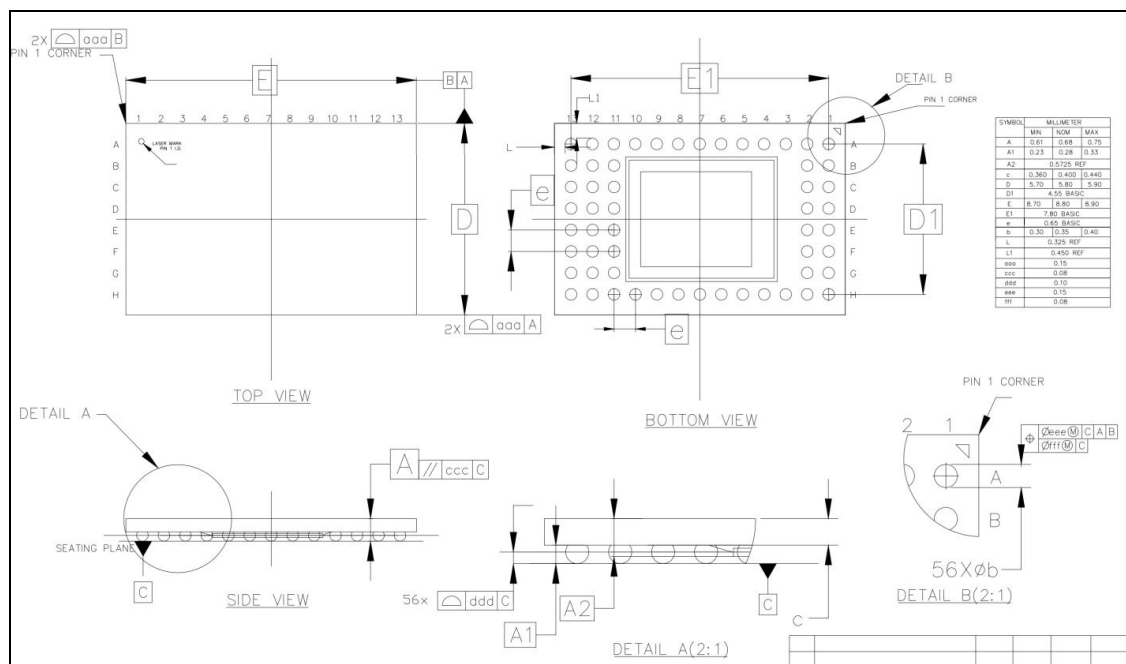


Figure 6-1 Packaging block diagram

6.2 PCB Pad Diagram

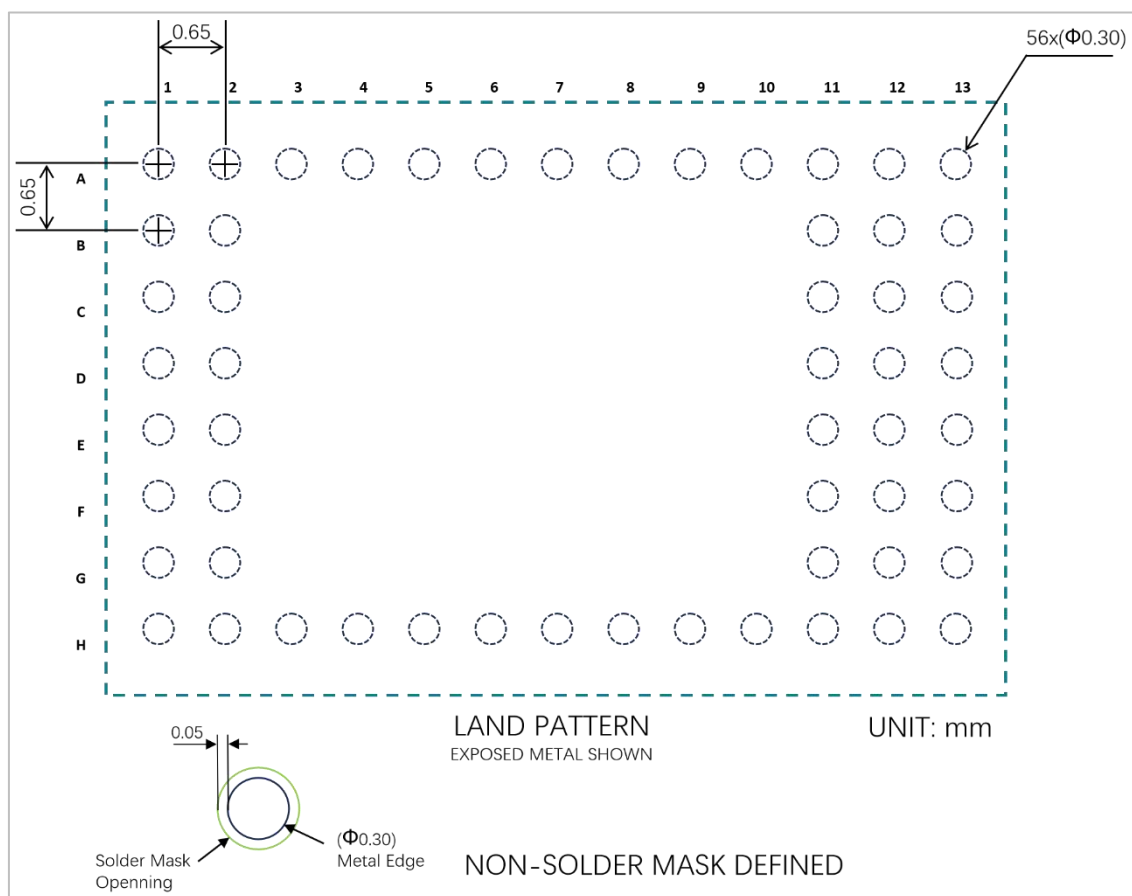


Figure 6-2 PCB pad diagram

7 Reference Schematic

7.1 Precautions

- **power supply**
 - **OPTION A:** Compared to using the RS6130 internal LDO to supply power to V13, using an **external** DCDC for power supply will result in more power savings. If the user is not sensitive to power consumption, this part of the circuit can be removed, thereby saving BOM costs.
 - VCC and VDDIO are powered externally. If VCC is lower than 3.6V, VCC and VDDIO can be tied together and powered by a single power supply.
 - VETX must be connected to the V18 pin and must not be connected to an external power supply or load.
- **Bluetooth**
 - **OPTION B:** "32.768KHz" is used for the Bluetooth subsystem. If the user does not require Bluetooth functionality, this part of the circuit can be removed.
 - **OPTION C:** "BLE ANTENNA" is used for the Bluetooth subsystem. If the user does not require Bluetooth functionality, this part of the circuit can be removed.

- **External connection**

The main controller can control the radar through any of the interfaces UART0, I2C0, or SPI0, with UART0 being the default.

The trigger output signal can be selected from PA6/PA11, and generally only one trigger signal is required.

If there is a need for OTA firmware, PB0 can be brought out to control the chip to enter the programming mode. For details, refer to Section 4.3.

To facilitate radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during noise floor testing.

- **OPTION D: CONNECT By I2C0**
- **OPTION E: CONNECT By SPI0**
- **For unused pins, leave them unconnected.**

7.2 Reference Schematic

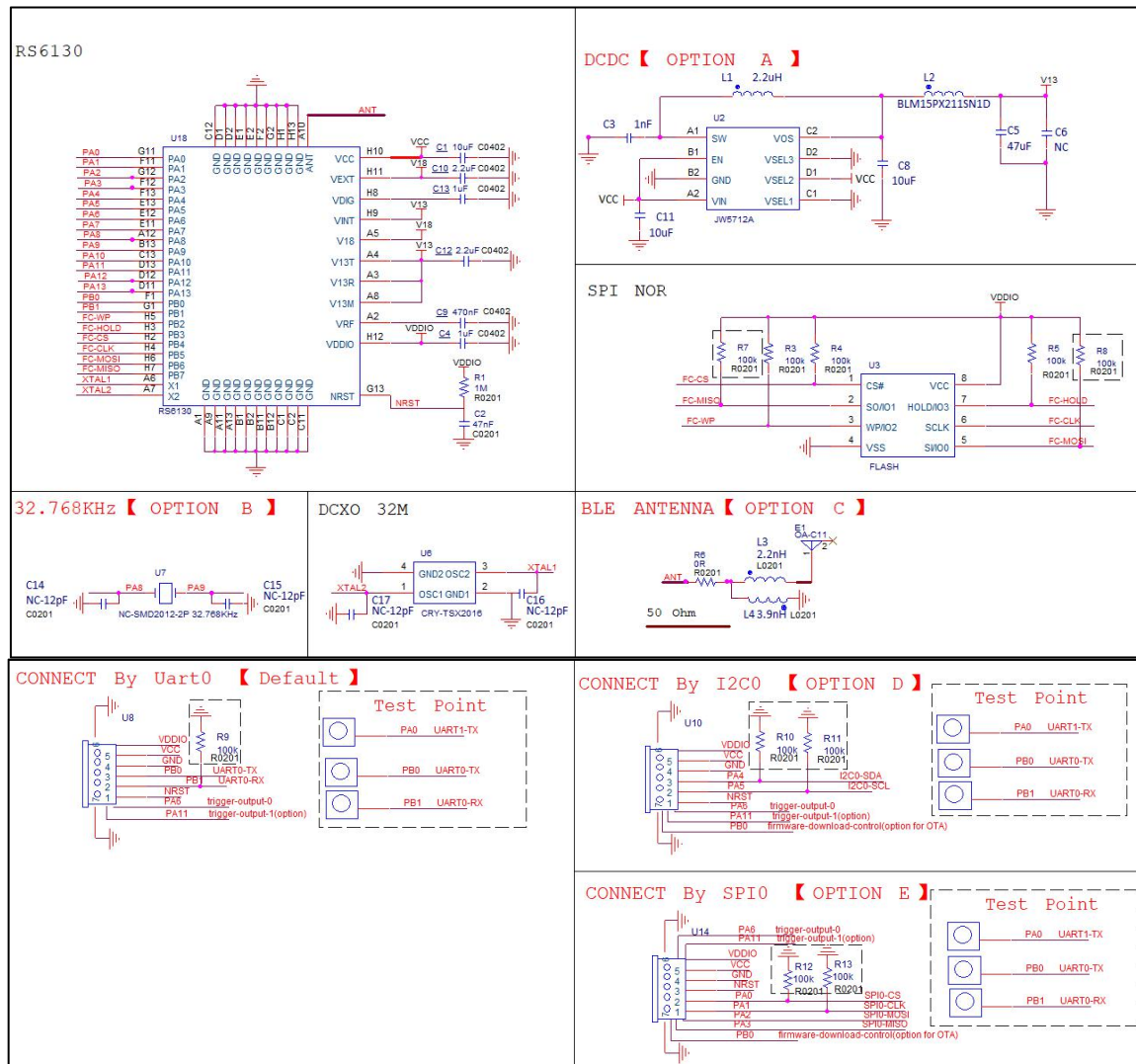


Figure 7-1 Reference schematic diagram

When designing the circuit, please carefully read the descriptions of each circuit module in the table below:

Table 7-1 Circuit Module Description

module	device	description
RS6130	C1,C4,C9,C10,C12,C13	Chip decoupling capacitor (X5R, accuracy $\pm 10\%$, voltage rating 6.3V), do not change the capacitance value.
	R1,C2	The NRST power-on RC delay ensures that the FLASH is ready before NRST is released. The parameters need to be set according to the power-on timing sequence of the flash. For details, refer to Section 5.1 Power-on Timing Sequence Description.
DCXO 32M	U6,C16,C17	The crystal specification is 32MHz, 12pF, with an accuracy of $\pm 10\text{ppm}$; C16 and C17 (C0G, accuracy $\pm 1\%$, voltage resistance 6.3V) are matching capacitors. The chip has built-in capacitors, so NC is the default setting.
32.768KHz [OPTION B]	U7,C14,C15	The low-frequency crystal oscillator operates at 32.768KHz with an accuracy of $\pm 20\text{ppm}$. C14 and C15 (C0G, with an accuracy of $\pm 1\%$ and a voltage resistance of 6.3V) serve as matching capacitors, which cannot be in the non-conducting (NC) state. The capacitance values are determined based on the selected model. This crystal oscillator provides precise timing for the Bluetooth system in standby mode. If Bluetooth is not used, it can be removed.
BLE ANTENNA [OPTION C]	R6,L3,L4,E1	Bluetooth antenna and matching circuit; the matching circuit needs to be modified according to the selected antenna type, and the antenna feed signal trace needs to be impedance-matched to 50 ohms; if Bluetooth is not used, it can be deleted, and the signal can be left idle.
SPI NOR	R3,R4,R5,R7,R8,U3	R3, R4, R5, R7, and R8 are pull-up resistors (accuracy $\pm 5\%$); U3 is a flash memory, and its selection should be made from the "Key BOM Selection Guide Manual". If WINBOND flash is chosen for U3, R7 and R8 need to be added for pull-up to enable the flash to enter the lowest sleep power consumption mode; when using flash from other brands, R7 and R8 can be removed.
DCDC [OPTION A]	Selection description	Compared to using the RS6240 internal LDO to power V13, using an external DCDC for power supply results in greater power consumption savings. If the user is not sensitive to power consumption, this part of the circuit can be removed, thereby saving BOM costs.
	L1,C3,C8,C11	JW5712A typical matching circuit L1: DCDC matching inductor, $\text{DCR} \leq 300\text{m}\Omega$, rated current $\geq 300\text{mA}$, such as LQM18PN2R2MFRL C: Matching capacitor (X5R, accuracy $\pm 10\%$, voltage resistance 6.3V)
	U2	JW5712A, a low-power DCDC, is not recommended for replacement with other models; if replacement with other models is necessary, they must meet the requirements of a quiescent current less than 600nA, an operating current greater than 400mA, and an output voltage of 1.3V. For more models, please refer to the "Key BOM Selection Guide Manual".
	L2	DCDC output filter bead, $220\Omega @ 100\text{MHz}$, $\text{DCR} \leq 0.1\Omega$, rated current $\geq 250\text{mA}$, such as BLM15PX221SN1D.

	C5,C6,C7	DCDC output filter storage capacitor (X5R, accuracy $\pm 20\%$, voltage resistance 6.3V), with varying capacitance values suitable for low-power applications with different power supply types: ➤ Batteries with high internal resistance such as button batteries and dry batteries: Add a 200uF capacitor at the battery terminal, reserve 2 capacitor positions for standby, and attach 47uF capacitors to C5, C6, and C7. If the actual layout space is insufficient, the capacitance value can be appropriately reduced, and refer to "Section 9.2. DCDC Output Performance Measurement" to confirm performance. ➤ Lithium batteries or batteries with low internal resistance such as capacitors: Reserve 2 capacitor positions at the battery terminal for standby use. Attach 10uF capacitors to C5 and C6, and leave C7 as NC (no connection).
CONNECT By UART0 [Default]		➤ The main controller can control the radar through UART0/I2C0/SPI0, with UART0 as the default. For ease of radar performance analysis, it is strongly recommended to reserve PA0, PB0, and PB1 as test points, as these three signals will be used during PCB performance analysis.
CONNECT By I2C0 [OPTION D]		
CONNECT By SPI0 [OPTION E]		➤ VCC and VDDIO are powered externally. If VCC is lower than 3.6V, VCC and VDDIO can be tied together and powered by a single power supply. ➤ PA6/PA11 is typically used as an interrupt trigger signal to wake up the host. ➤ If there is a need for online firmware updates, PB0 needs to be pulled out to control the chip to enter programming mode. ➤ If VCC is not used as a power switch, the NRST signal needs to be connected to the main controller to assist in implementing abnormal reset or updating firmware online. ➤ For the pins RX (PB1) of UART0, CS (PA0) and CLK (PA1) of SPI0, and SDA (PA4) and SCL (PA5) of I2C0, they cannot be in a floating state when the radar is in hibernation, otherwise the radar may experience abnormal leakage. If the main control terminal cannot set the corresponding signal pins to a fixed level during hibernation, external pull-down resistors R9/R10, R11/R12, and R13 need to be added to fix the level. If the main control terminal can set the level, they can be removed.

7.3 SPI Multi-radar Hardware Design

7.3.1 SPI Multi-radar Usage Constraints

When using a multi-radar solution, users must ensure that the multi-radar can perform normal programming, which requires meeting the following constraints:

- During the SPI multi-radar programming process, the unselected radar VDDIO must be in a powered state.
- During the SPI multi-radar programming process, the unselected radars must be in a reset state or application mode.

7.3.2 Recommended Solution

Based on the above constraints, our company offers the following multi-radar hardware structure scheme:

1. The host controls the NRST pins of different radars individually through the IO port. During programming, for radars not selected by SPI_CS, NRST is in a pulled-down state, ensuring that the unselected radars are in a reset state.
2. The VCC and VDDIO of each radar module are connected in parallel and independently controlled through switches. During programming, all radars are powered on to ensure that VDDIO is powered.

Note: The radar programming shall be performed in accordance with the specified procedure. For detailed information, refer to Section 3.1.2 in the RS6xxx_Firmware Download_Development Manual.

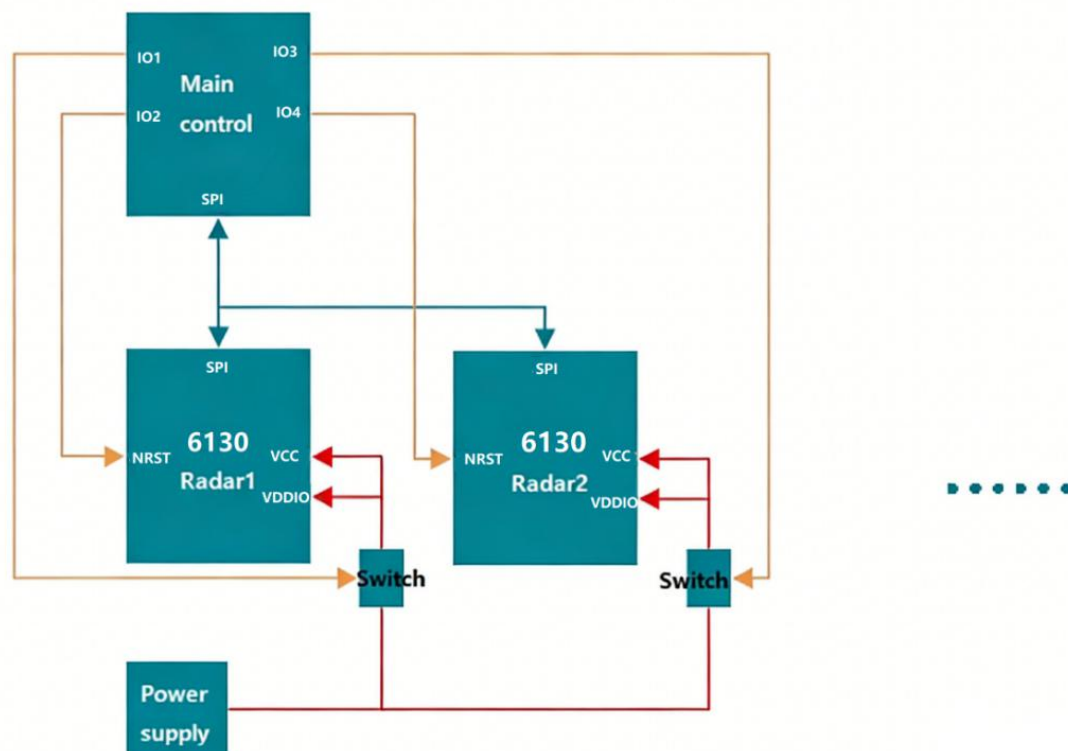


Figure 7–2 Schematic diagram of SPI multi-slave hardware connection

If designed according to the aforementioned scheme, in the event of an anomaly in a certain radar module, the module can be powered on again through an independently controlled power switch to perform a complete reset, without affecting other modules.

If the number of IO ports on the user's host device is insufficient, the following simplified solution can be used, as shown in Figure 7-3. All power switches are integrated into a single power bus switch. The limitation of this solution is that all radar modules must be powered on and off synchronously.

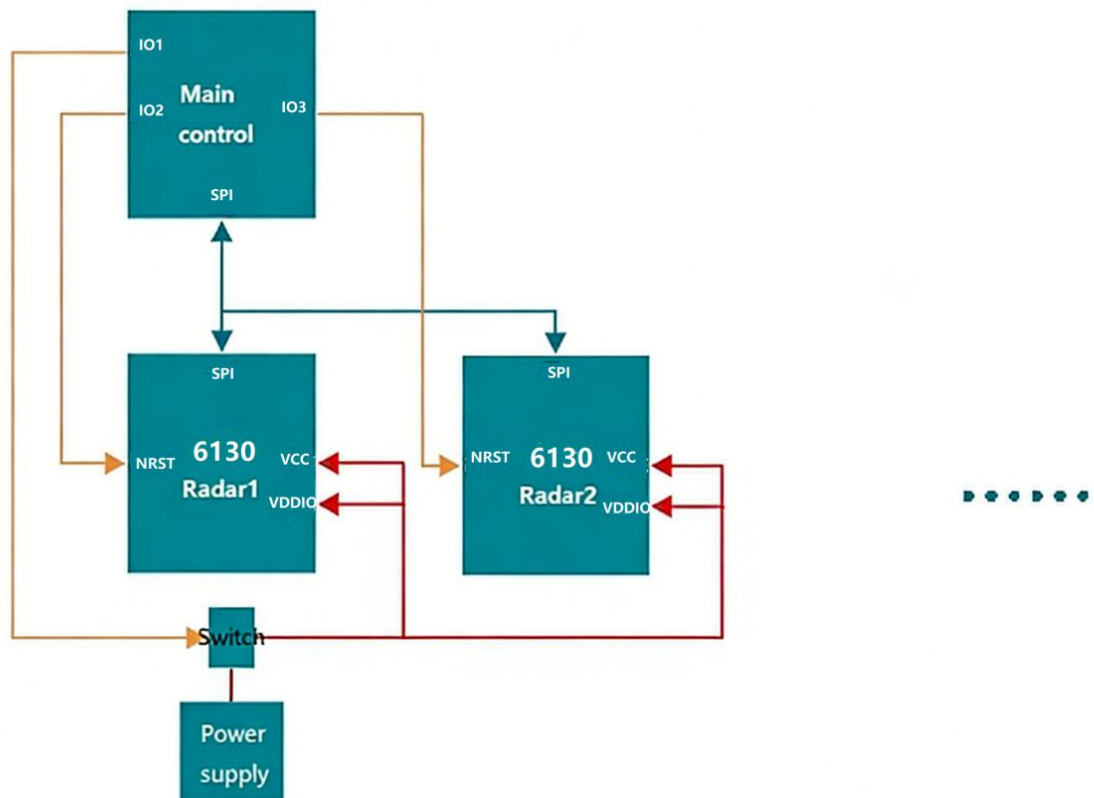


Figure 7-3 Single power switch scheme for SPI multi-slave hardware structure

8 PCB Layout Guide

The performance of RS6130 radar is sensitive to component and GND layout. For different application requirements, it is divided into PCB single-sided layout and PCB double-sided layout.

PCB single-sided layout is suitable for applications where radar angle and directional pattern requirements are not stringent. It is recommended not to use DCDC circuits for boards smaller than 25mm*25mm.

The double-sided PCB layout requires the radar chip to be placed on a separate side, which is suitable for applications with high requirements for radar angle and directional pattern.

It is recommended to adopt a four-layer board design for the radar hardware application solution, and the following will provide layout design guidelines based on a four-layer board.

8.1 PCB Ground Plane Optimization

In addition to the radome support structure and the surrounding environment of the radar, the size of the PCB floor also has a certain impact on the radar's performance. Therefore, if necessary, special treatment can be applied to the PCB to meet the requirements for superior radar performance.

Below are the results of the radiation patterns of RS6130 under different PCB board sizes:

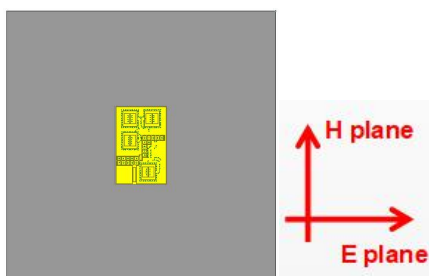
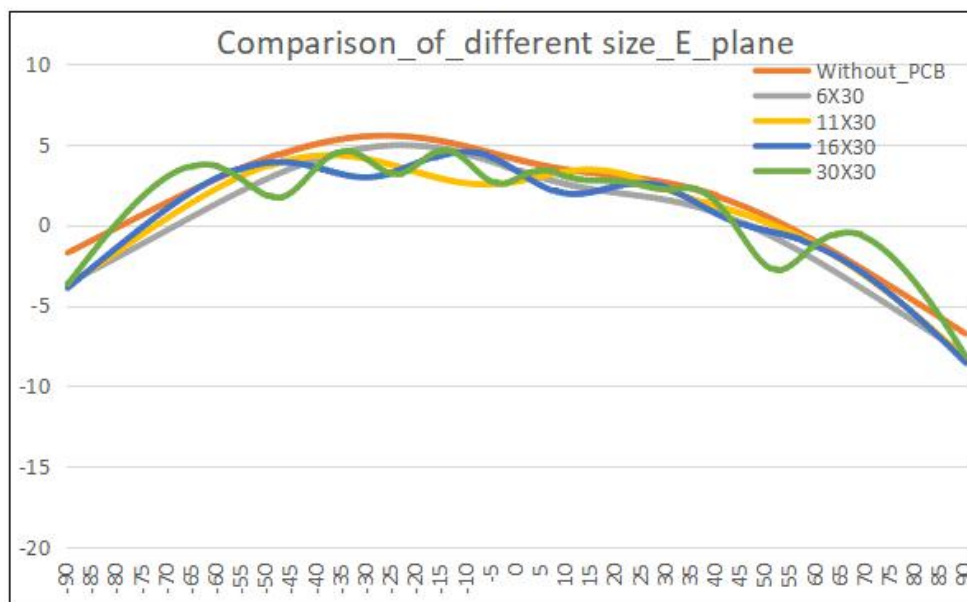


Figure 8–1 Schematic diagram of PCB floor direction definition



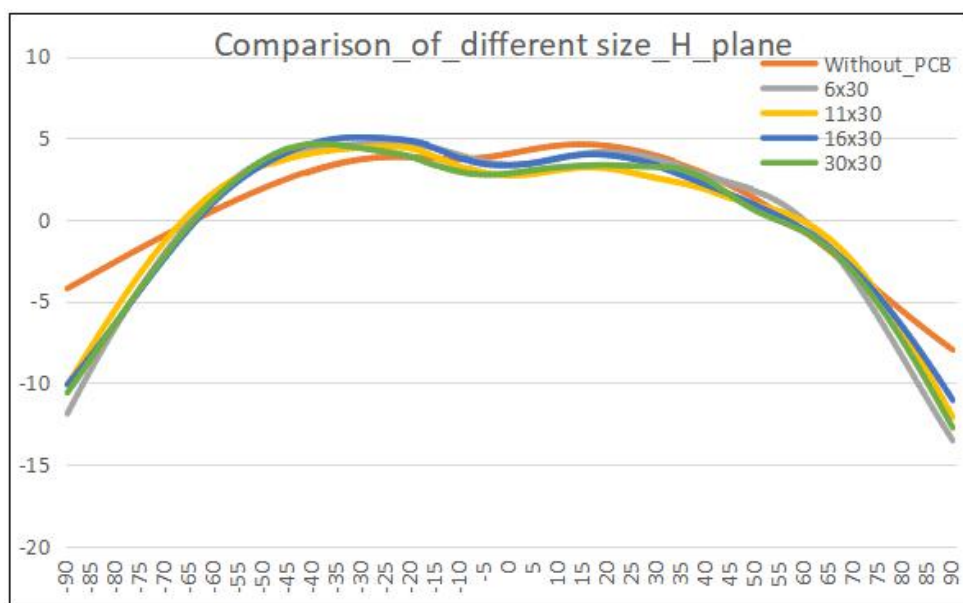
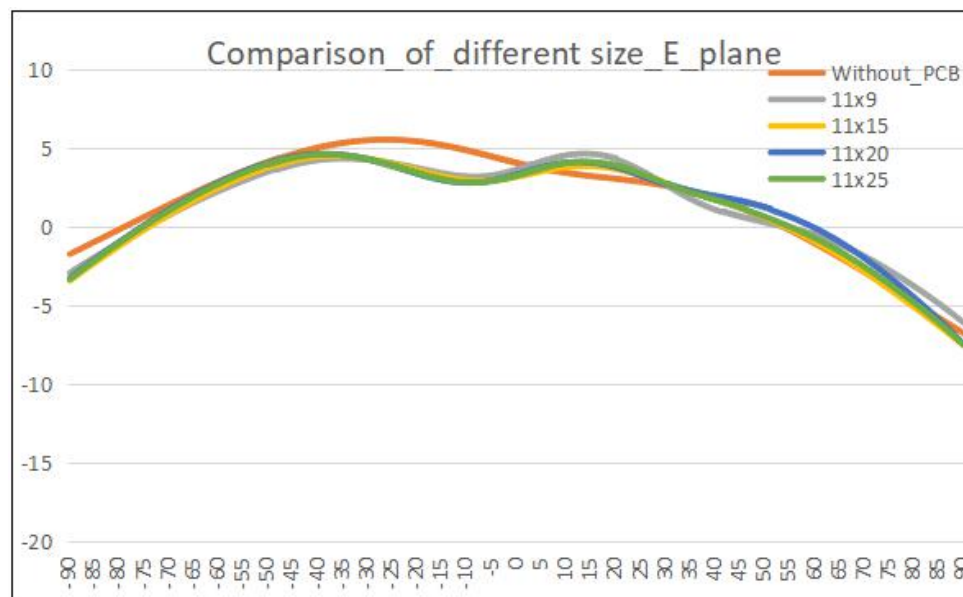


Figure 8–2 Comparison of radiation patterns of PCB along the E-plane dimension variation



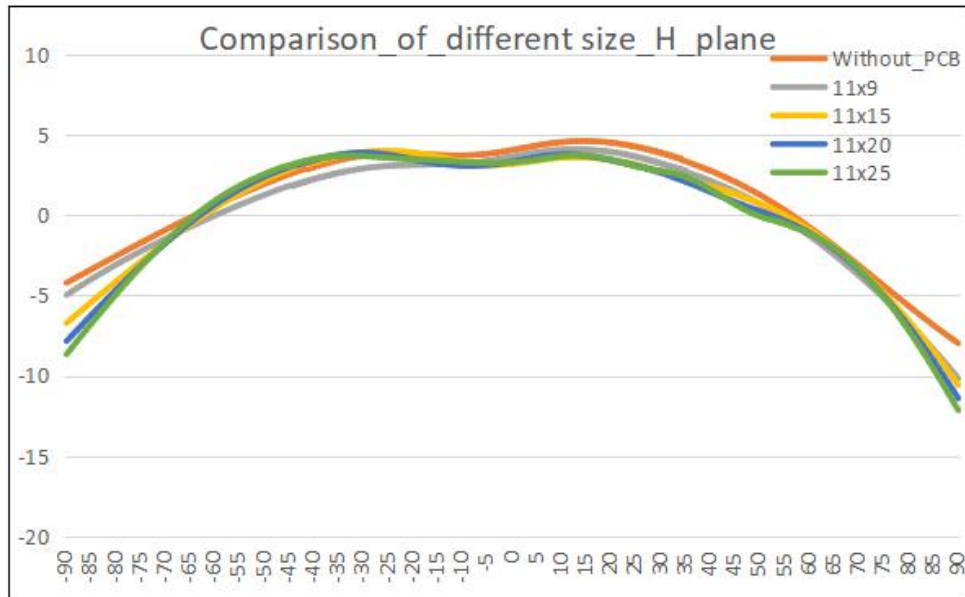


Figure 8–3 Comparison of radiation patterns along the H-plane dimension variation of PCB

According to Figure 8–3, the radiation pattern of RS6130 in the H plane basically does not change significantly with the increase in PCB size, while the radiation pattern in the E plane exhibits ripples as the PCB size increases. For every additional air wavelength between the PCB edge and the RS6130 substrate edge, a ripple is introduced. After reaching a certain level, the radiation pattern at large angles deteriorates more severely due to the influence of ripples, which is caused by the surface waves on the PCB floor.

To mitigate the impact of PCB flooring on radar performance, an optimized design can be implemented specifically for the PCB flooring.

Option 1: Limit the PCB size in the E-plane direction or perform copper removal on the PCB to ensure that the distance from the PCB board edge or copper-clad area to the edge of the chip package is greater than half a wavelength and not exceeding one wavelength, that is, $2.5\text{mm} \leq d \leq 5\text{mm}$, as shown in the figure below:

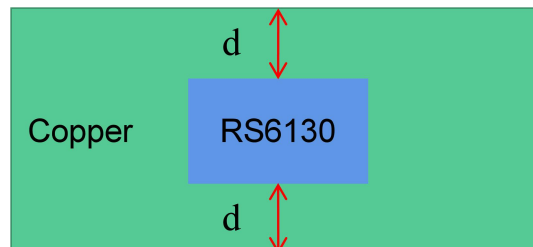


Figure 8–4 Schematic diagram of copper-clad area in Scheme 1

Option 2: Perform trapezoidal windowing on the PCB, with the trapezoidal windowing being spaced $d=5\text{mm}$ from the chip. The height of the trapezoid, $h\lambda_0 = 5\text{mm}$, should be $\geq 3/4$ of the air wavelength, i.e., $h \geq 3.75\text{mm}$, and $w \geq 3.75\text{mm}$. Other dimensions are as shown in the figure below:

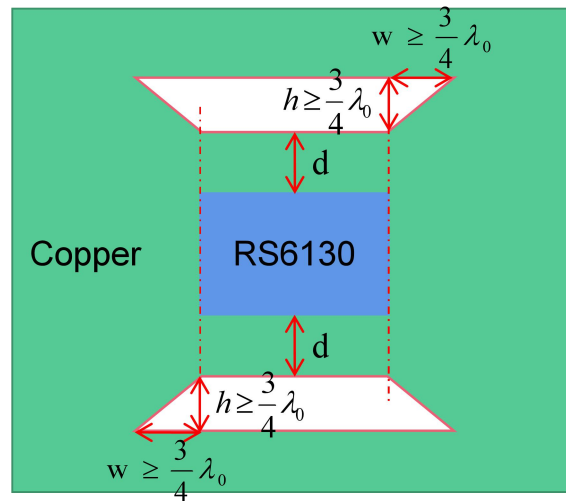
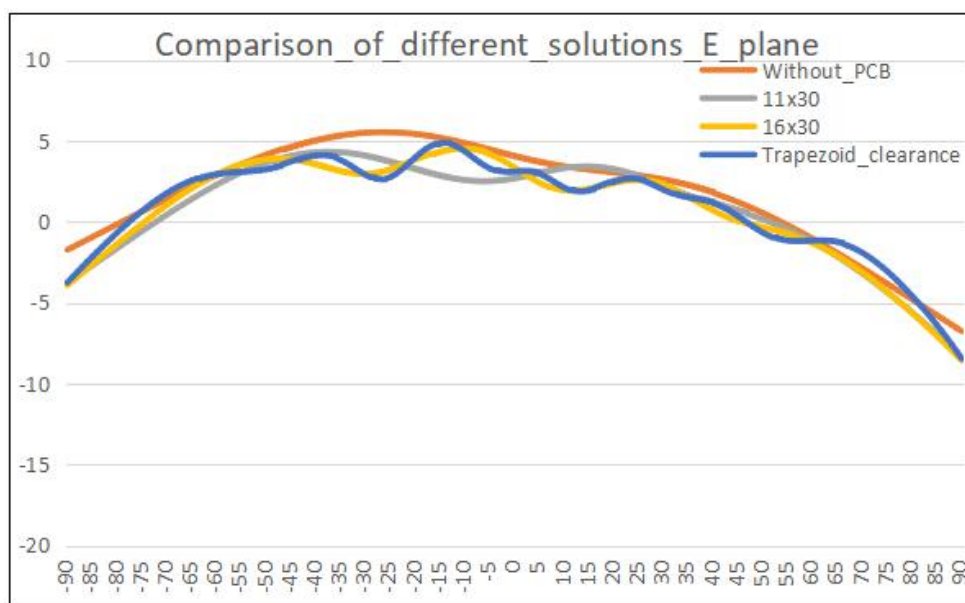
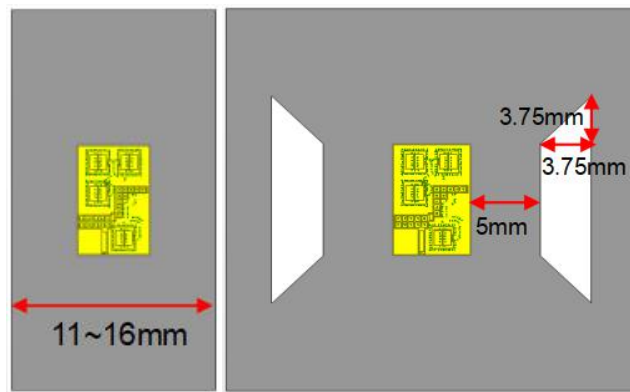


Figure 8–5 Schematic diagram of trapezoidal window opening in Scheme 2

After undergoing copper removal treatment and trapezoidal windowing, the simulation results of its radiation pattern are compared as follows:



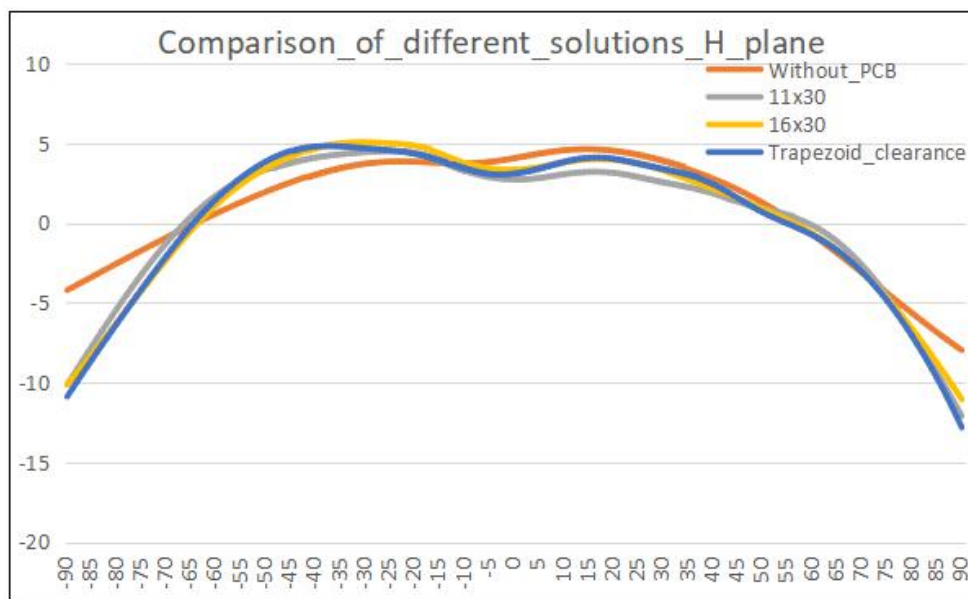
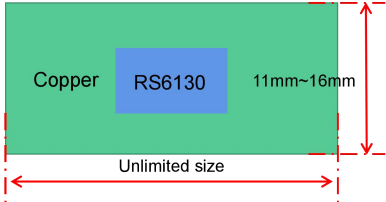
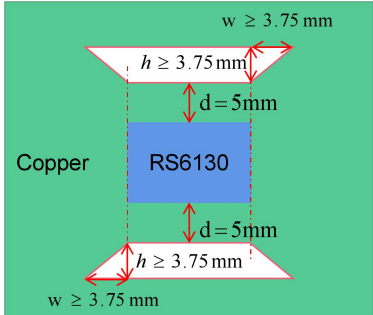


Figure 8–6 Comparison of radiation patterns between Scheme 1 and Scheme 2

According to Figure 8-6, there is no restriction on the PCB floor size in the H-plane direction. However, in the E-plane direction, to ensure optimal performance, the PCB floor size in the E-plane direction can be selected between 11mm and 16mm. When the PCB size exceeds 16mm, the two aforementioned solutions need to be adopted.

Table 8-1 Reference design for millimeter-wave radar PCB floor

chip type	RS6130
PCB Type 1	 No requirements for PCB on the long side of the chip PCB size of chip short edge: 11mm~16mm
PCB Type 2	 There is no requirement for the PCB size of the long side of the chip In cases where the short side of the chip PCB is greater than 16mm, a trapezoidal window shall be added to the PCB floor. The distance from the window to the edge of the chip shall be d=5mm, and the length of the parallel short side of the trapezoid shall be consistent with the length of the chip's long side h≥3.75mm, w≥3.75mm

8.2 Chip Placement Orientation

Due to the characteristics of the chip antenna, the radiation intensity in different directions may vary slightly. When no antenna cover is applied, the FOV (Field of View) in the E-plane is approximately 130 degrees, and in the H-plane, it is about 110 degrees. When a flat antenna cover is applied, the FOV in the E-plane is about 120 degrees, and in the H-plane, it is about 100 degrees. Therefore, if the usage scenario involves side mounting, it is recommended to place the chip vertically and use the E-plane to detect horizontal directions. As shown in the figure below.

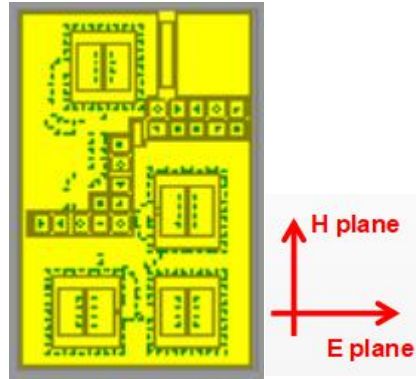


Figure 8-7 Chip placement orientation

8.3 PCB Single-sided Layout Requirements

For the top-level device layout on one side, the chip should be placed in the middle of the board in a symmetrical position. Capacitors such as VRF, V13, VCC, VDDIO, V18, and VDIG must be placed close to the side of the chip, as shown in the following figure:

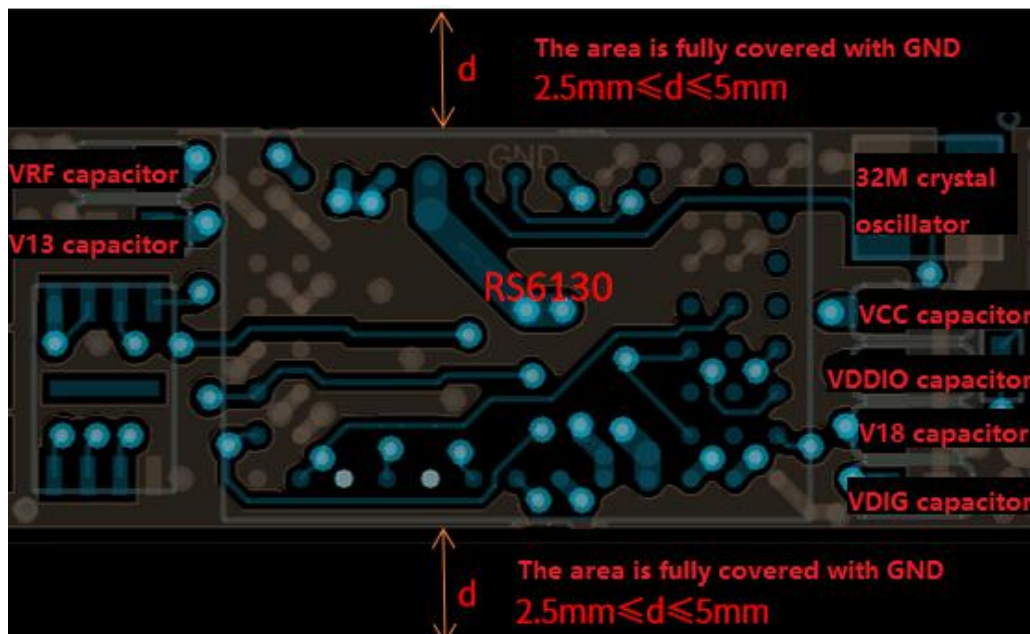


Figure 8-8 Top layer of PCB single-sided layout (radar chip side)

For the single-sided layout, the second layer should be fully covered with GND, as shown in the following figure:

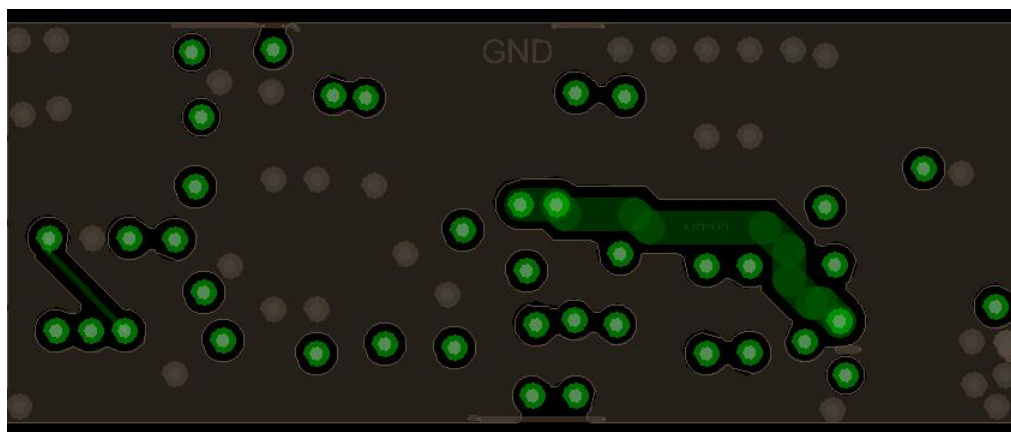


Figure 8-9 PCB single-sided layout, second layer with GND

Single-sided layout with DCDC design, refer to the following figure for Layout:

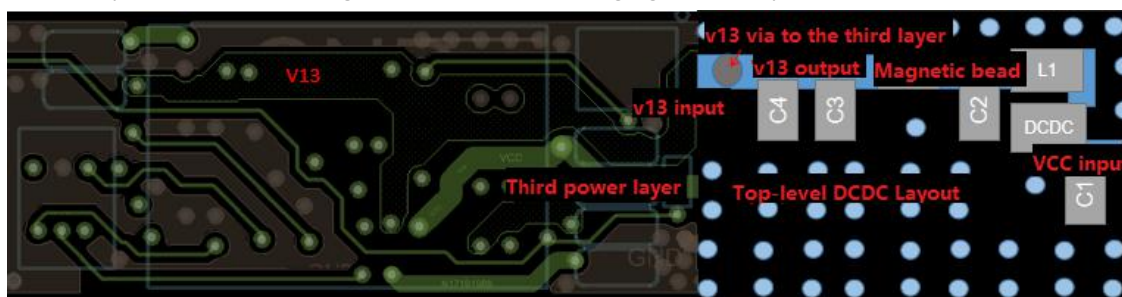


Figure 8-10 Reference diagram for PCB single-sided layout of DCDC Layout

Requirements for single-sided layout:

- The decoupling capacitors for VCC, VDDIO, VRF, V18, and V13 are situated close to the radar chip, with a spacing of less than 2mm;
- No wiring is allowed in areas outside the top-level chip to ensure the continuity and integrity of the GND;
- The power line width requirements are that the line widths of VCC and V13 must be greater than 0.5mm, and the line widths of VDDIO, V18, and VRF must be greater than 0.25mm;
- V13 and V18 are power-sensitive, and their routing should not overlap with signal lines and VDDIO on adjacent layers.
- Refer to Section 8.3 for DCDC Layout.

8.4 PCB Double-sided Layout Requirements

For the dual-sided layout, only radar chips are placed on the top layer, and these radar chips are arranged symmetrically in the middle of the board, as shown below:

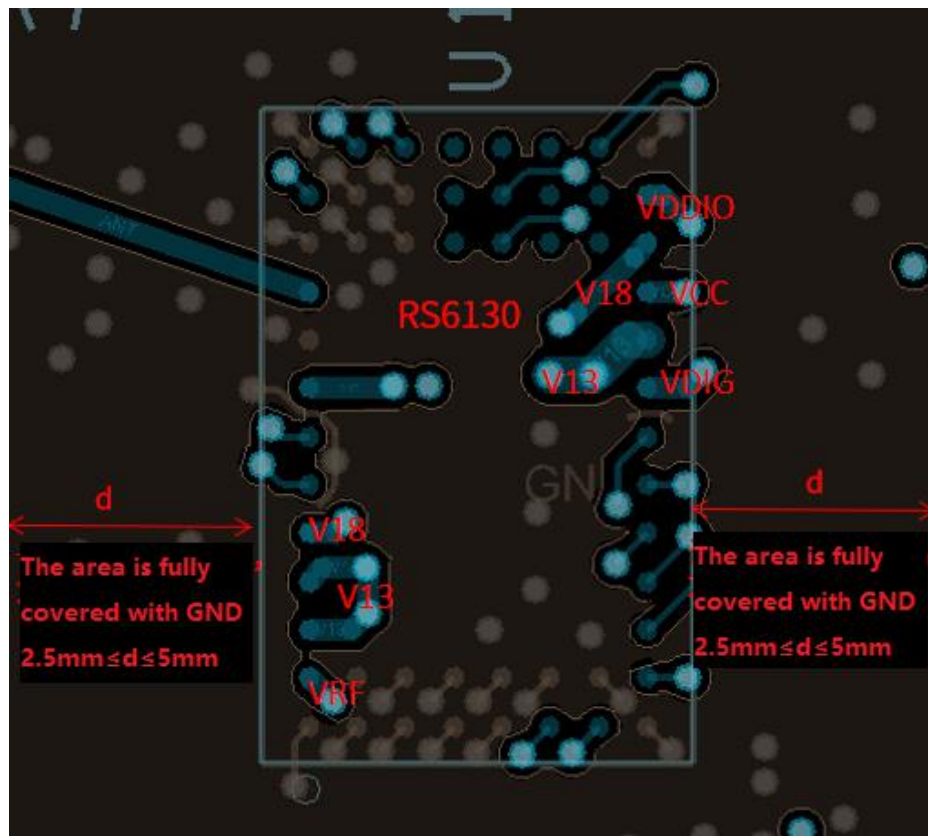


Figure 8-11 Top-level Layout of PCB Double-sided Layout

The second layer of the double-sided layout is completely covered with GND, as shown below:

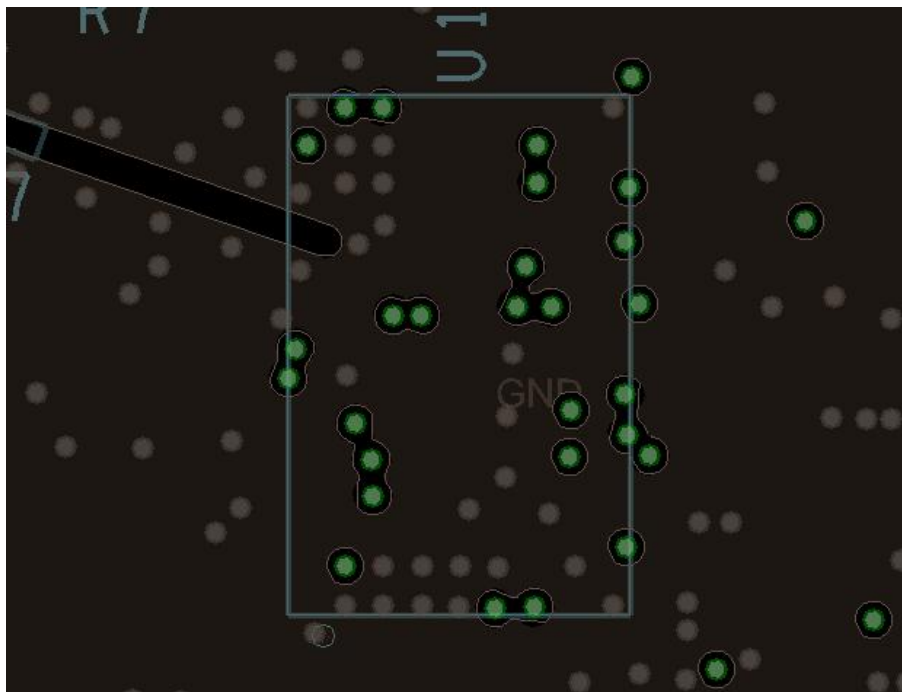


Figure 8-12 PCB double-sided layout, second layer with GND

For the fourth-layer device layout in double-sided layout, capacitors for VRF, V13, VCC, VDDIO, and V18 must be placed closely adjacent to the chip pins. There are two ways to connect the DCDC output to V13, as shown in Figures 8-13 and 8-14:

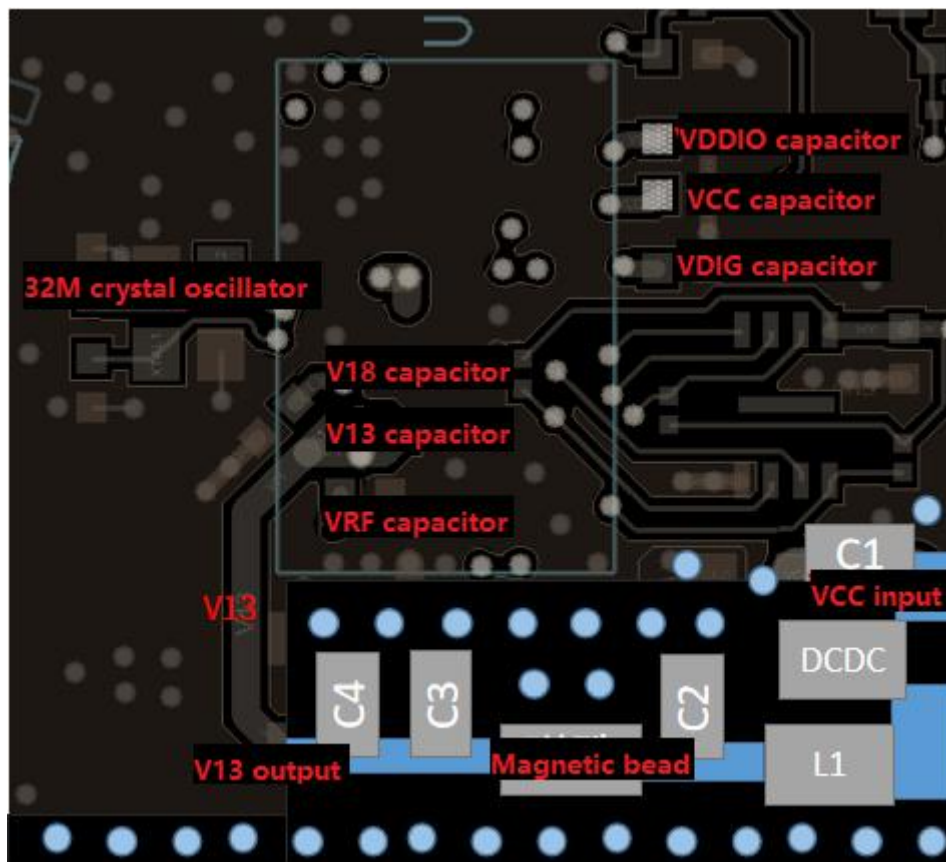


Figure 8-13 PCB double-sided layout, fourth layer Layout (DCDC layout 1)

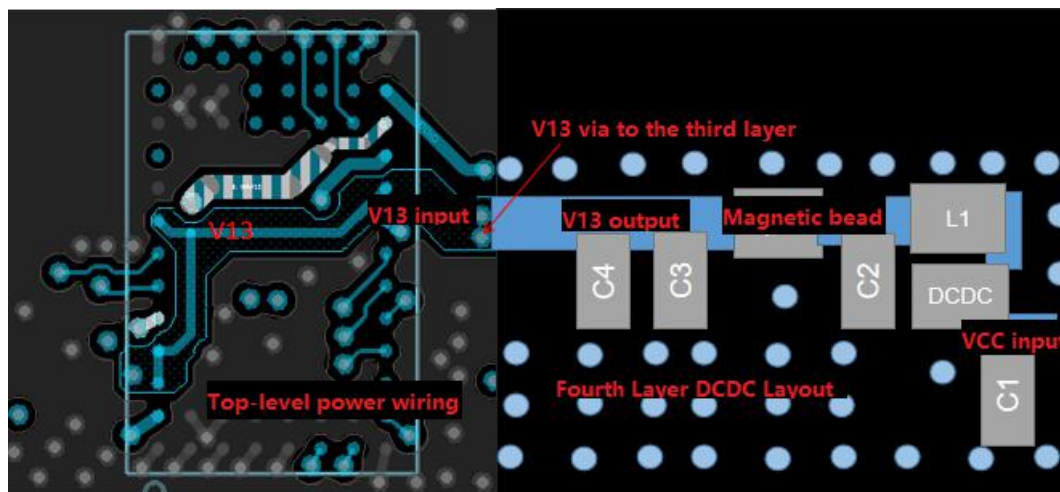


Figure 8-14 PCB double-sided layout DCDC Layout (DCDC Layout 2)

PCB double-sided layout requirements:

- The decoupling capacitors for VCC, VDDIO, VRF, V18, and V13 are placed close to the radar chip pins, with a spacing within 2mm;
- The signal and power lines are routed through vias in the shaded area of the chip to the third and fourth layers;

- The power line width requirements are as follows: the line widths of VCC and V13 must be greater than 0.5mm, while those of VDDIO, V18, and VRF must be greater than 0.25mm;

8.5 DCDC Layout

Schematic diagram of DCDC matching circuit layout:

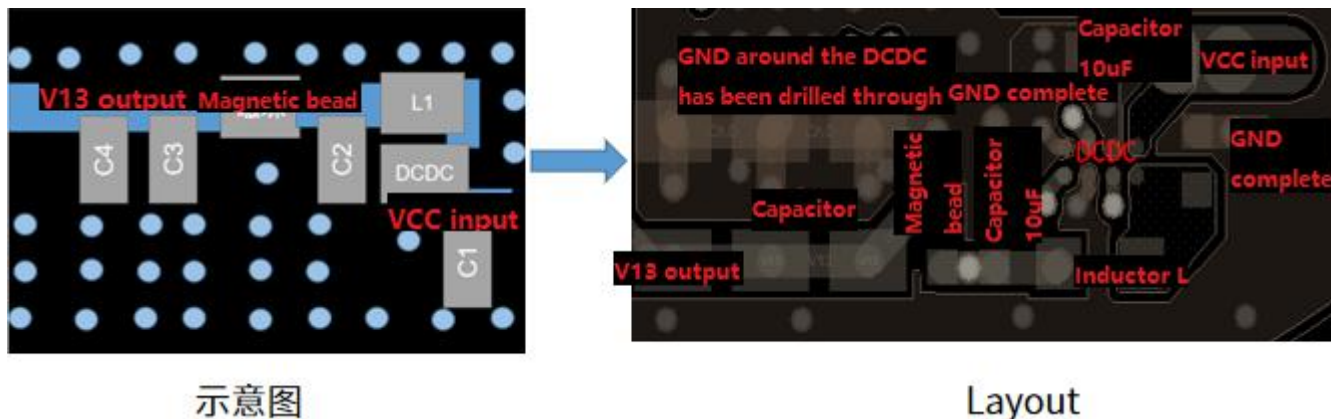


Figure 8-15 DCDC Layout

DCDC circuit Layout requirements:

- The DCDC matching circuit is laid out according to the layout shown in the figure above;
- The surrounding GND of the DCDC must be complete, and the surrounding GND has been drilled;
- When DCDC and matching circuits are laid out on the same layer as the radar chip, they must be separated by at least 5mm. When they are laid out on different layers, they must be separated by at least 1mm, and the separation area should have a complete and continuous GND.

8.6 Crystal Oscillator Layout

Top-level layout schematic diagram of 32M crystal oscillator clock:

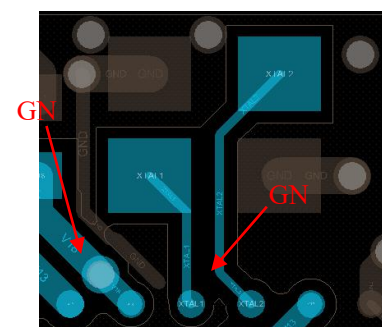


Figure 8-16 Clock layout

Layout requirements for 32M crystal oscillator clock:

- The 32M crystal oscillator should be isolated from power supplies V13 and V18 by GND, with traces kept as short as possible;
- Lay GND on both sides of the X1 and X2 signal traces to reduce unnecessary signal coupling;
- The crystal oscillator GND via connects to the second layer GND.

8.7 Antenna Layout

Schematic diagram of BLE antenna top-level layout:



Figure 8-17 BLE antenna layout

Precautions for BLE antenna layout:

- The antenna wiring should be as short as possible;
- The impedance of antenna wiring must be 50Ω , and impedance control should refer to the third-layer GND;
- Lay GND near the antenna wiring;
- The GND vias on both sides of the antenna are symmetrical;
- Add a π matching circuit.

8.8 External SPI Layout

If the user employs the SPI interface for communication with an external host, the following points should be noted:

- The trace impedance is controlled at 50 ohms (with a tolerance of $\pm 10\%$) for single-ended signals.
- The requirement for equal length between the clock and data is less than 5mm.
- The length of the wire is less than 10cm.
- The spacing between signal lines should be at least twice the width of the lines.

8.9 Layout Example - P1806 Module (No BLE/No DCDC/Single-sided)

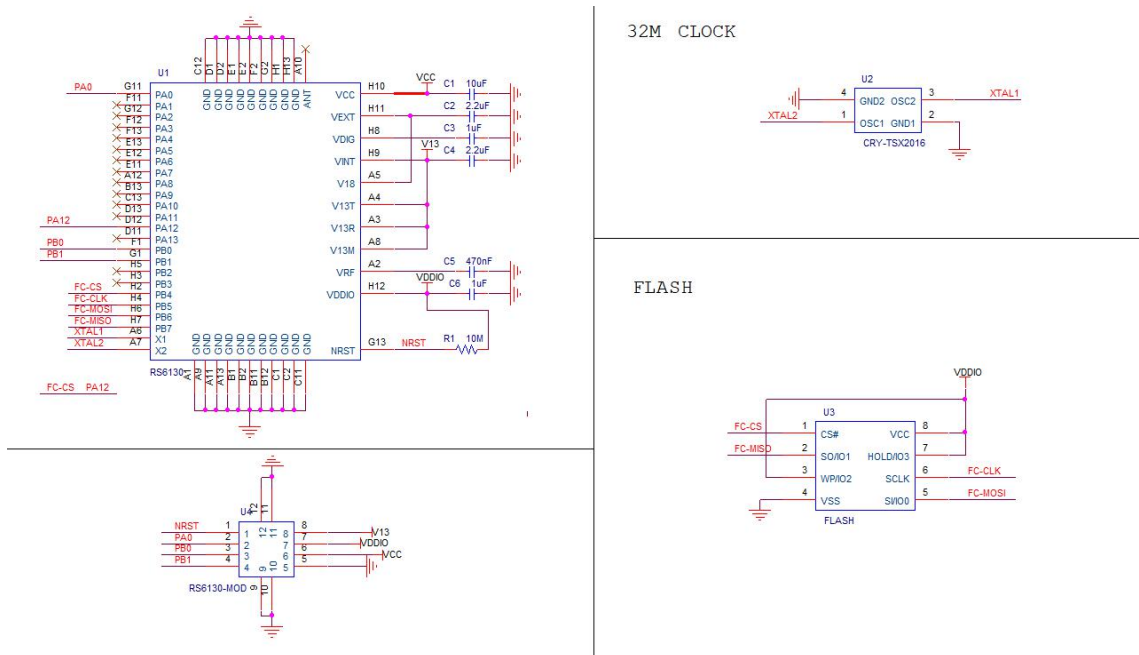


Figure 8-18 Schematic diagram of P1806 module

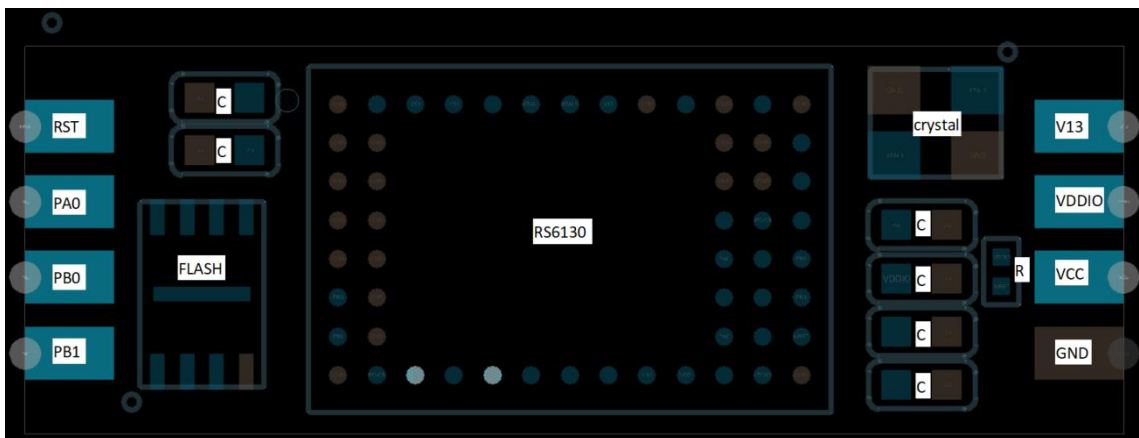


Figure 8-19 TOP layer device layout

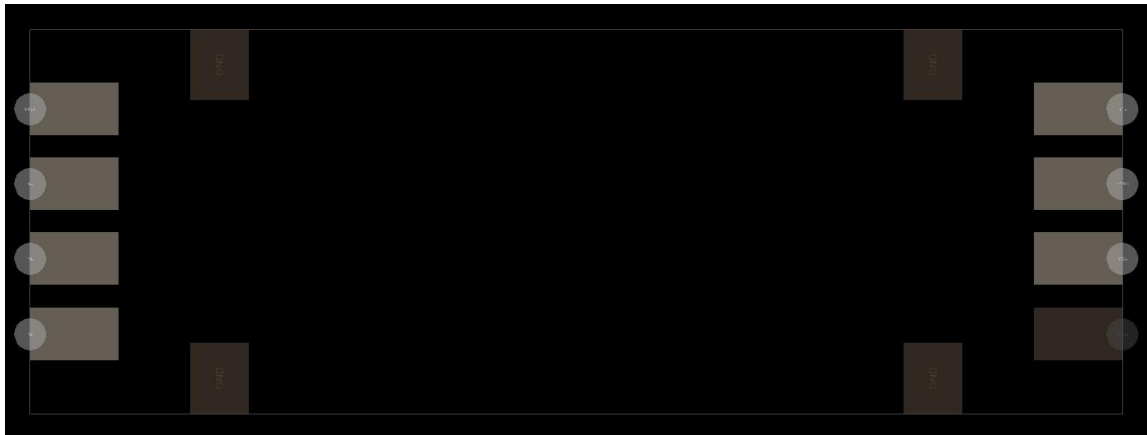


Figure 8-20 BOTTOM layer Overview

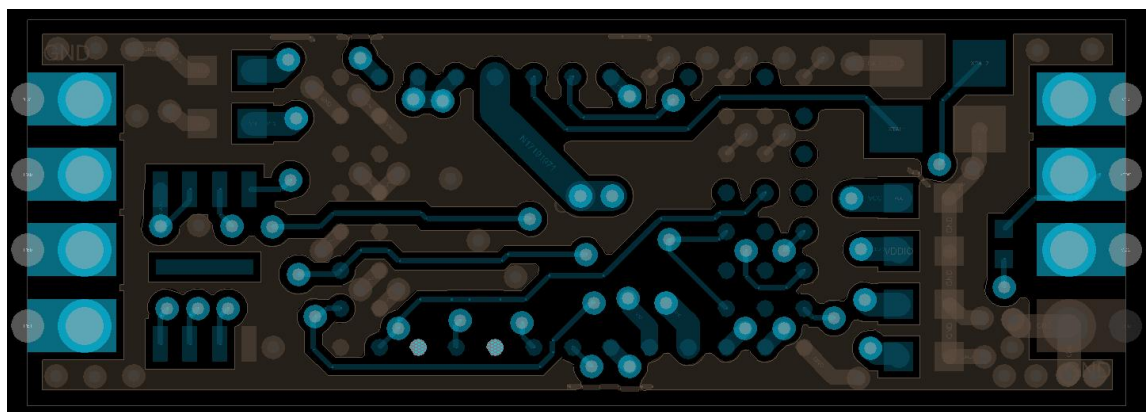


Figure 8-21 TOP layer layout

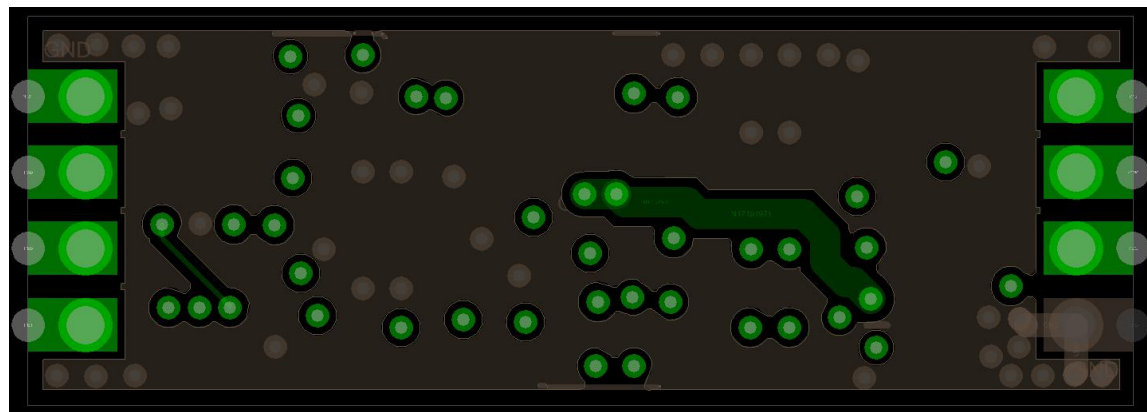


Figure 8-22 Layout of GND02 layer

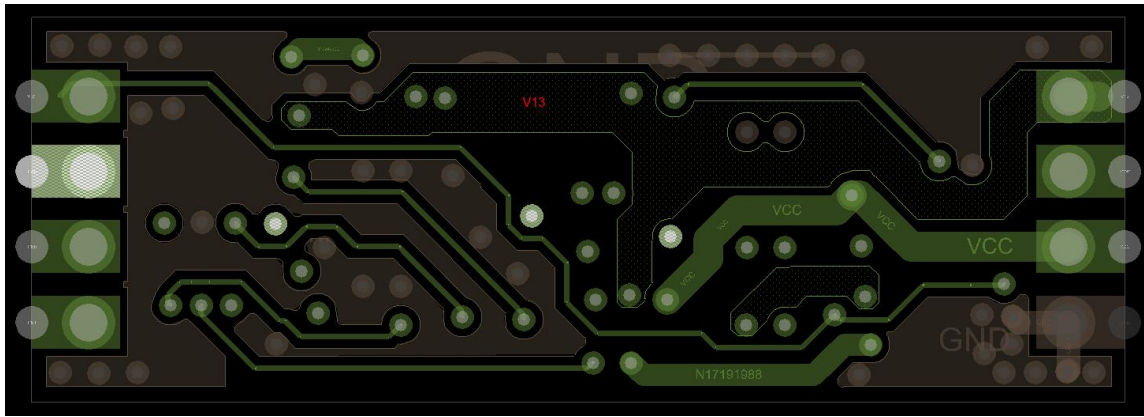


Figure 8-23 Layout of PWR03 layer

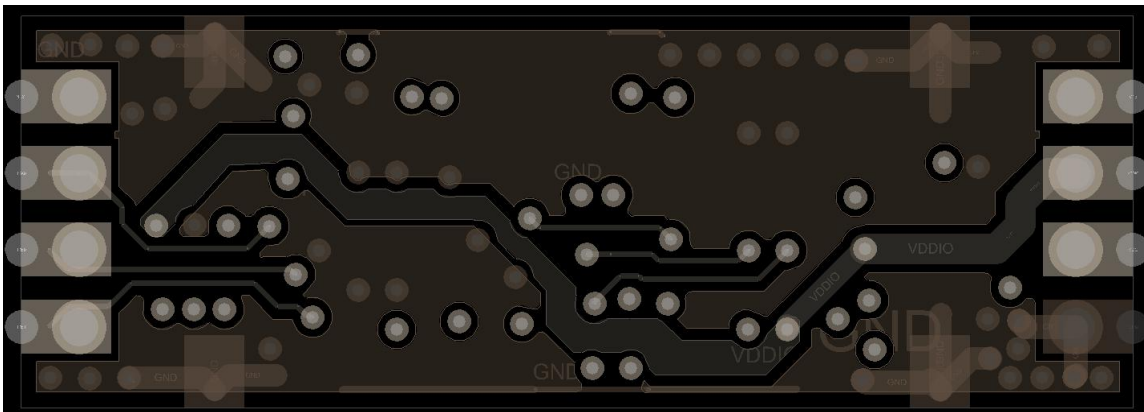
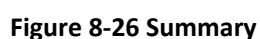


Figure 8-24 BOTTOM layer layout

If the PCB layout space is limited, users can refer to this layout design.



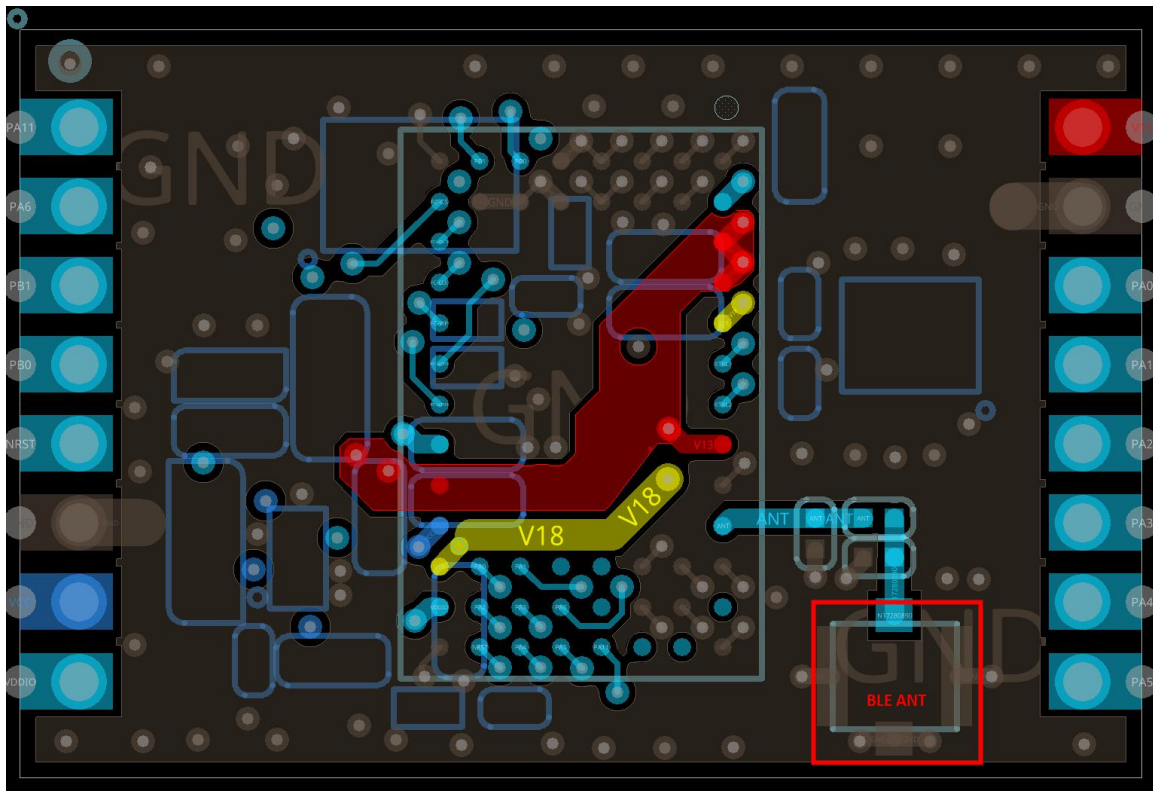


Figure 8-27 TOP layer LAYOUT

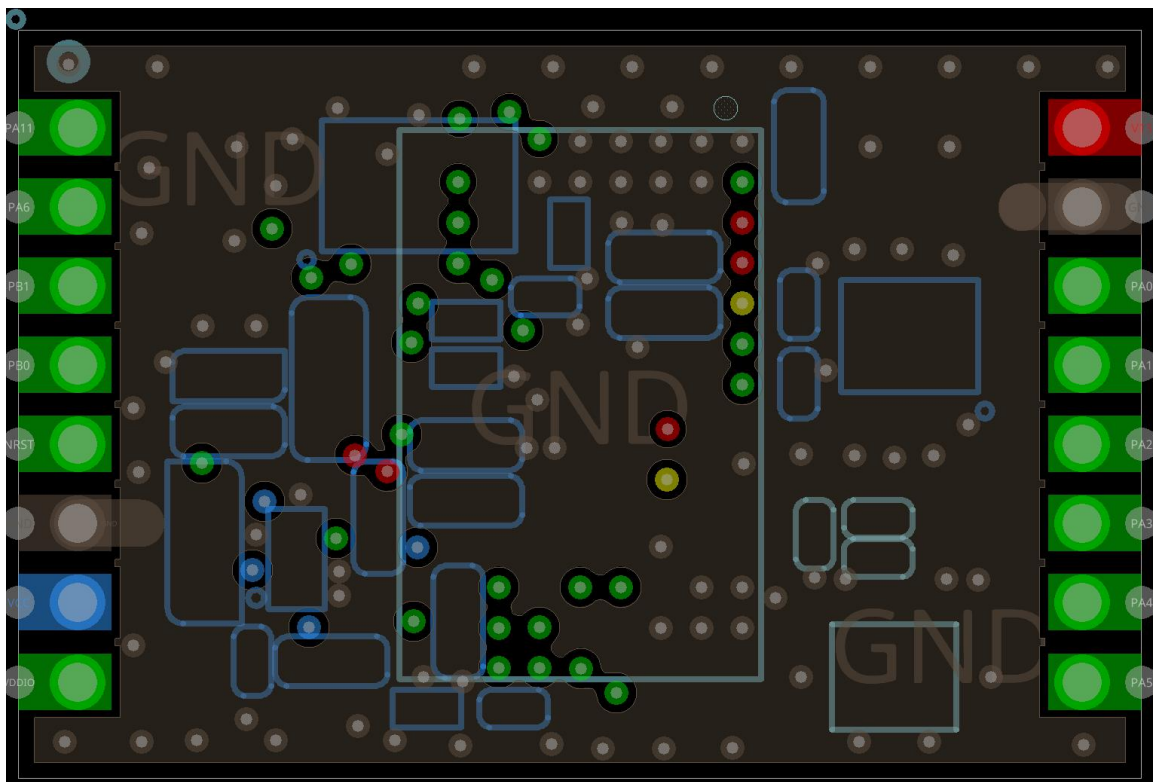


Figure 8-28 LAYOUT of GND02 layer

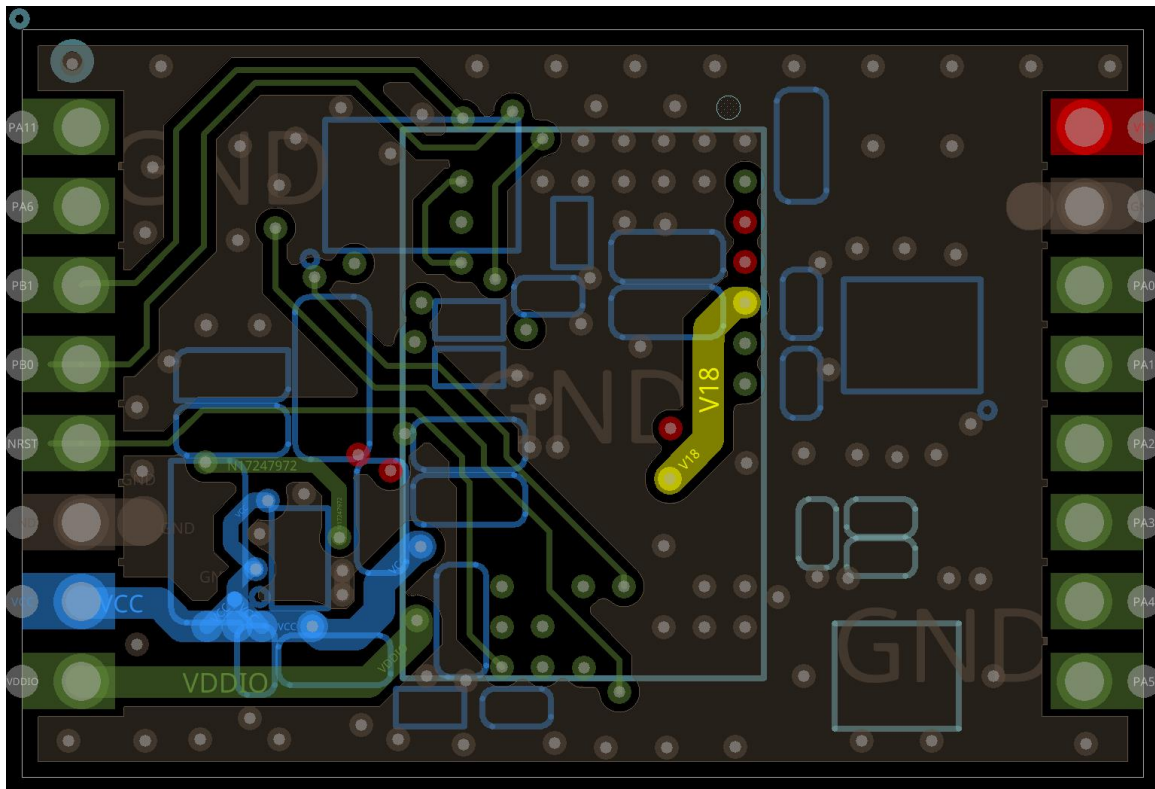


Figure 8-29 LAYOUT of PWR03 layer

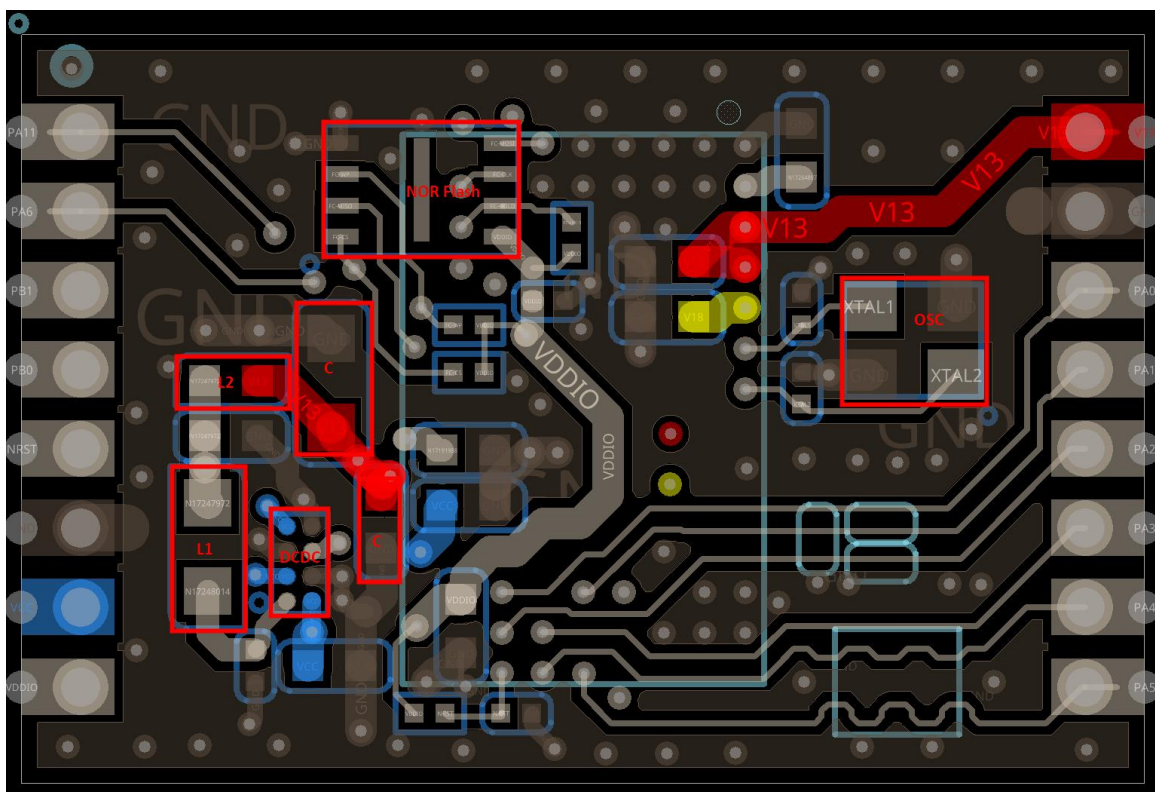


Figure 8-30 BOTTOM layer LAYOUT

9 PCB Performance Analysis

9.1 Noise Floor Measurement

Poor PCB design can introduce excessive noise into the radar system, thereby affecting ranging performance. The following will introduce methods for measuring noise, so that users can evaluate the quality of PCB design.

Since different schemes correspond to different software data packages, users need to select the appropriate noise floor measurement method based on the content of the obtained software package. There are two methods:

- ◆ Based on measurements from "RadarAnalysisTool" (for SDK development users)
- ◆ Based on measurements from "MotionDetectionDemoTool" (for users who do not require development).

Note: The host computer tool and dedicated test firmware can be obtained from the path [Resource Package\Software Kit\04_image](#).

9.1.1 Measurement with RadarAnalysisTool

The test method requires the use of PA0 (UART1-TX), PB0 (UART0-TX), and PB1 (UART0-RX). Therefore, during PCB design, it is necessary to ensure that these three signals can be brought out.

Test preparation:

- Host computer tool "RadarAnalysisTool"
- PCB to be tested (burned with dedicated test firmware according to the module model, either "radar_analysis_rs6130_p1806.img" or "radar_analysis_rs6130_p1812.img")
- Serial port adapter board (connecting UART0-TX, UART0-RX, UART1-TX of PCB)
- Test environment: Ensure that only the radar under test is operating in the surrounding area, as the operation of other radars may cause interference.

9.1.1.1 Usage of the Upper Computer

Step 1: Power on the PCB and connect the two serial ports mentioned above to the PC. Open the host computer tool.

- (1) Click on "Platform Configuration" on the page.
- (2) Select the hardware corresponding platform "MRS6130-1812".
- (3) Connect the serial port, select the COM port corresponding to UART0 for the control port, with a baud rate of 115200, and select the COM port corresponding to UART1 for the data port, with a baud rate of 1000000.
- (4) Click "Connect Device".



Figure 9-1 PCB Performance Test Step 1

Step 2:

- (1) Click on "Parameter Configuration" on the page.
- (2) Configure the following parameters in order: Multi-antenna mode (1T3R), starting frequency (58), maximum measurement distance (10), maximum measurement speed (10), distance resolution (4), speed resolution (64), frame structure (2DFFT), Interval Chirp N (1), number of frames (0), frame period (1183)
- (3) Click "Start".

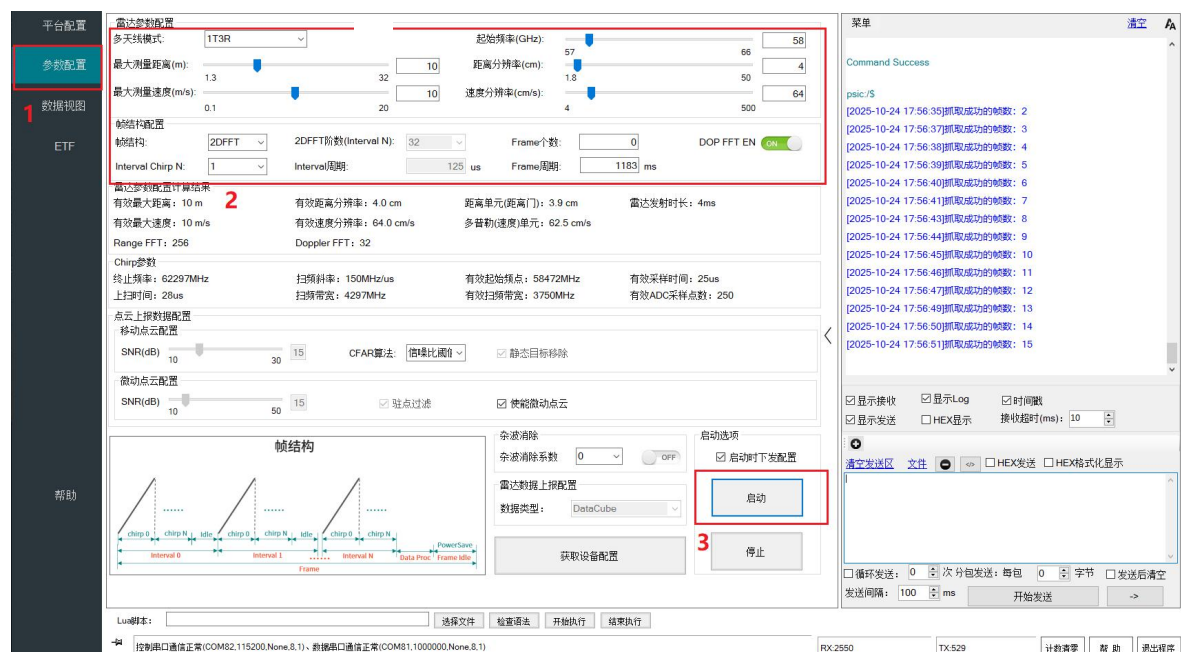


Figure 9-2 PCB Performance Test Step 2

Step 3:

- (1) Click on "Data View" on the page.
- (2) Click the "Image Display Mode" button and select "Single Image".
- (3) Check "Configuration" and uncheck "Highlight Max" in the pop-up menu.

- (4) Finally, set the static removal parameter to "10". Note that when setting the static removal parameter, you need to first select the original value with the mouse and then enter 10. The input will take effect immediately, and you can see the line chart as shown below.

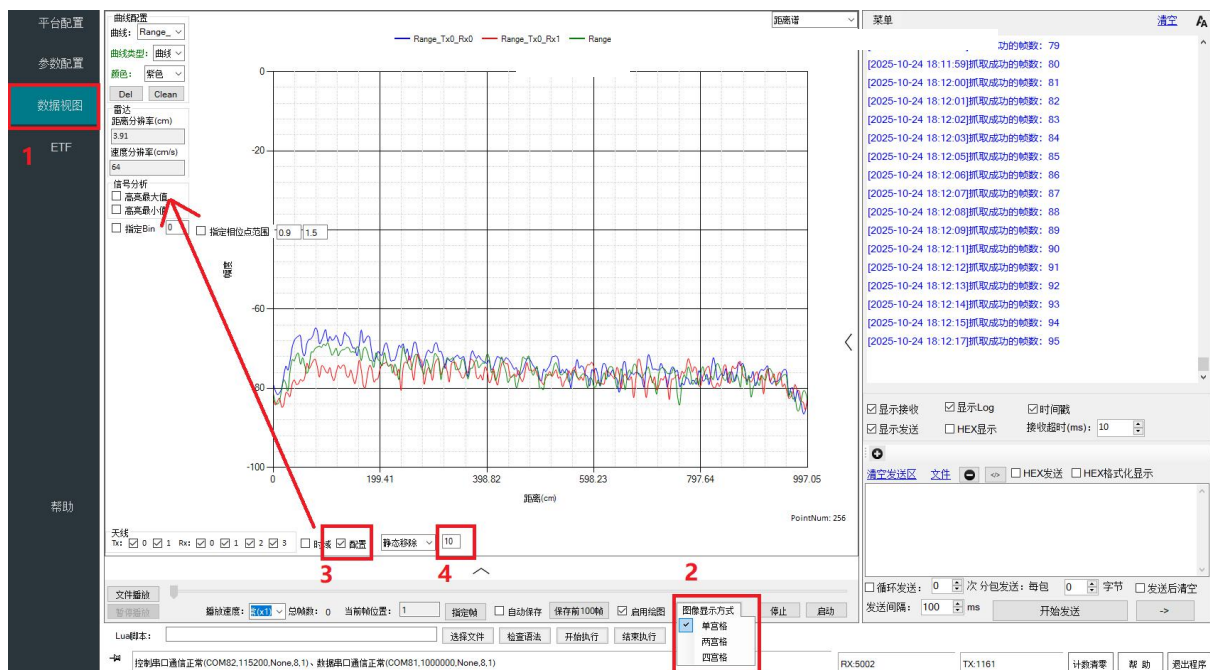


Figure 9-3 PCB Performance Test Step Three

9.1.1.2 Criteria for judgment

Under the condition of good hardware design, the measured noise characteristics are as follows:

- The overall amplitude center is approximately -76dB.
- The amplitude curve varies smoothly, with no obvious protruding peaks.

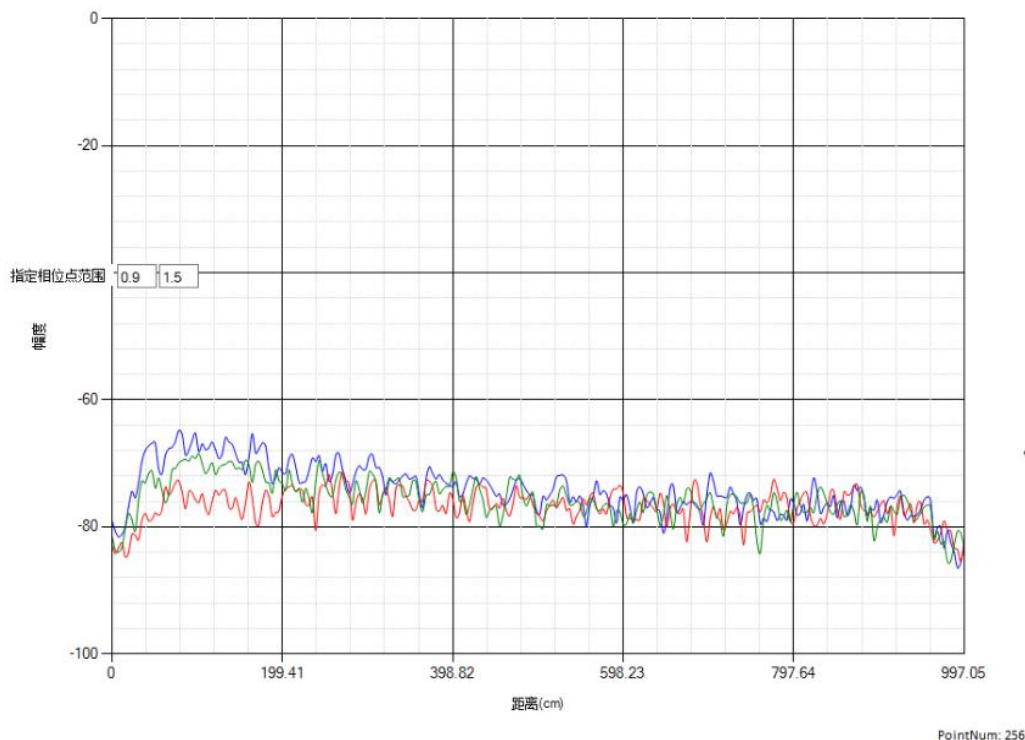


Figure 9-4 Normal graph

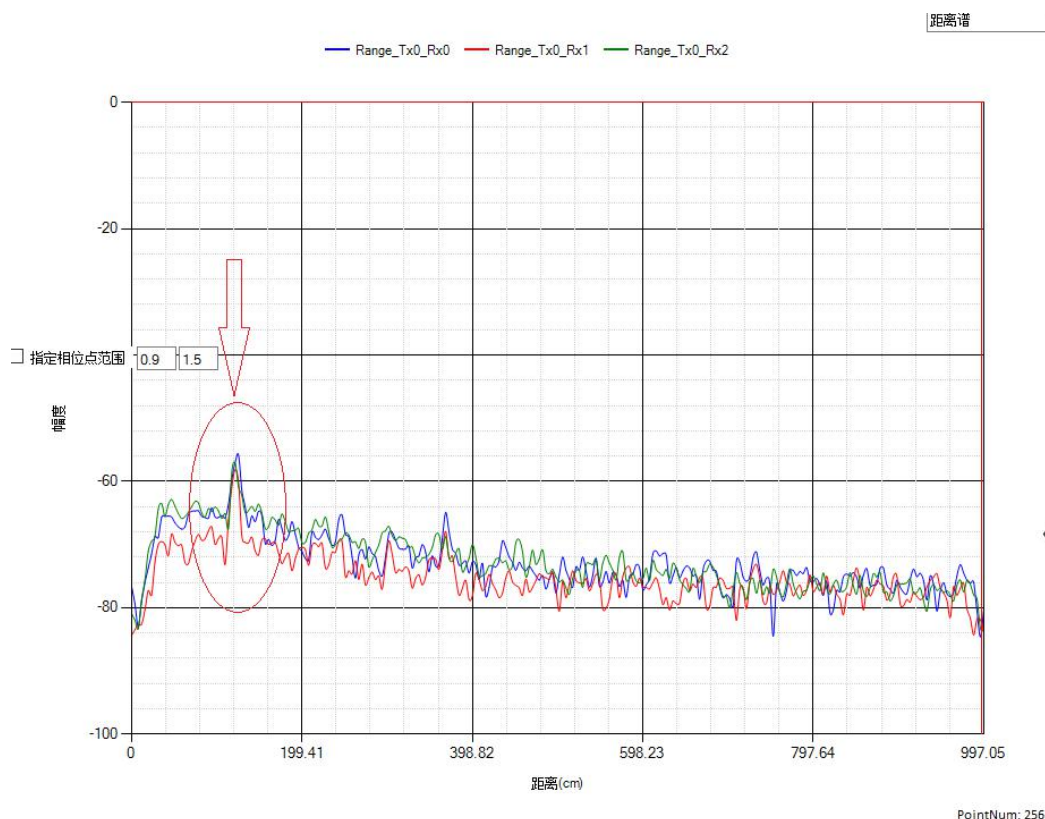


Figure 9-5 Abnormal Graph (Case of RS6130 Being Interfered by DCDC)

9.1.2 Measurement with MotionDetectionDemoTool

During the testing process, any of the UART/I2C/SPI interfaces can be used, depending on the communication interface chosen by the user during design. In actual testing, the PCB under test can generally be connected to the development board provided by our company, and then connected to the host computer. Therefore, when designing the PCB, it is necessary to ensure that the signals corresponding to the communication interface can be easily extracted and connected to the development board.

Test preparation:

- Host computer tool "MotionDetectionDemoTool"
- PCB to be tested (public firmware in the programming data package, refer to "RS6130_Mobile Detection DEMO_User Manual" for detailed programming instructions)
- Development board (MRS6130-AIP-DEV-V1)
- Test environment: Ensure that only the radar under test is operating in the surrounding area, as the operation of other radars may cause interference.

Note: The host computer tool and general firmware can be obtained from Resource [Package\Software Kit\04_image](#).

9.1.2.1 Usage of the upper computer

Power on the PCB and connect it to the PC via the development board. Open the host computer tool. **Within 5 seconds after the PCB is powered on, the radar will perform noise floor estimation. During this period, it is necessary to ensure that the environment in front of the radar is as static as possible, otherwise it will interfere with the noise floor estimation results.**

Step 1: After powering on for 5 seconds, select the corresponding communication interface and click "CONNECT".

Step 2: Check "Display Noise Floor" and click "Get Noise Floor" three times to obtain the noise floor data for the three antennas respectively.

Step 3: The background noise result will be displayed at the bottom right.

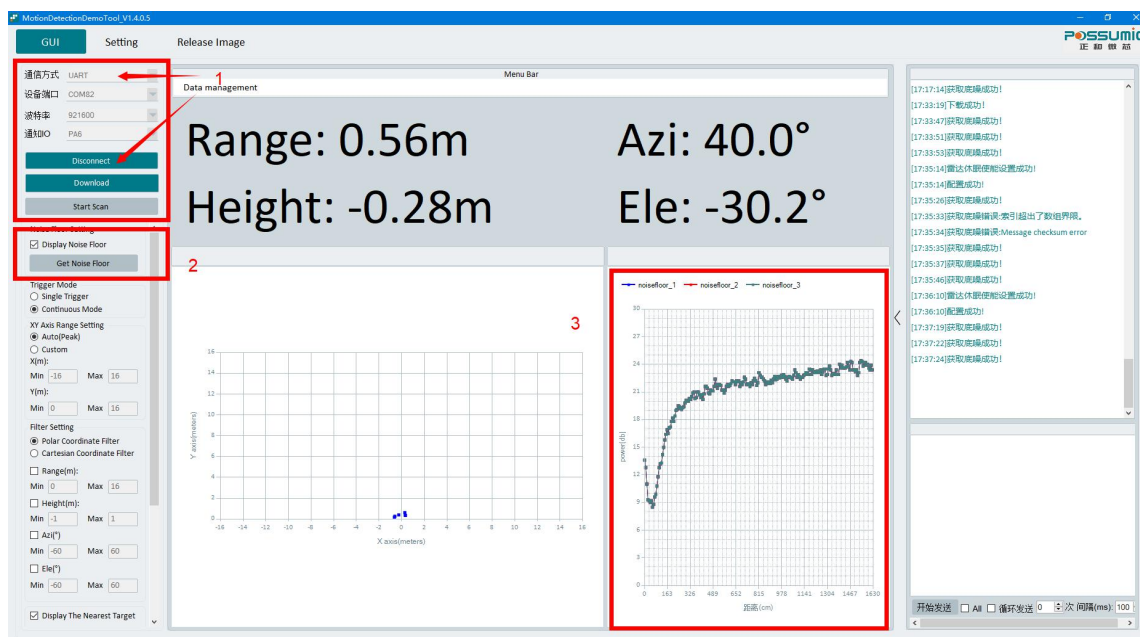


Figure 9-6 Setup steps for the host computer

9.1.2.2 Criteria for judgment

Under the condition of good hardware design, the measured noise characteristics are as follows:

- The background noise changes slowly, with no obvious prominent noise (noise characteristics: peak value is 3~4dB higher than the nearby values).
- After multiple measurements, the observed results all meet the aforementioned requirements.

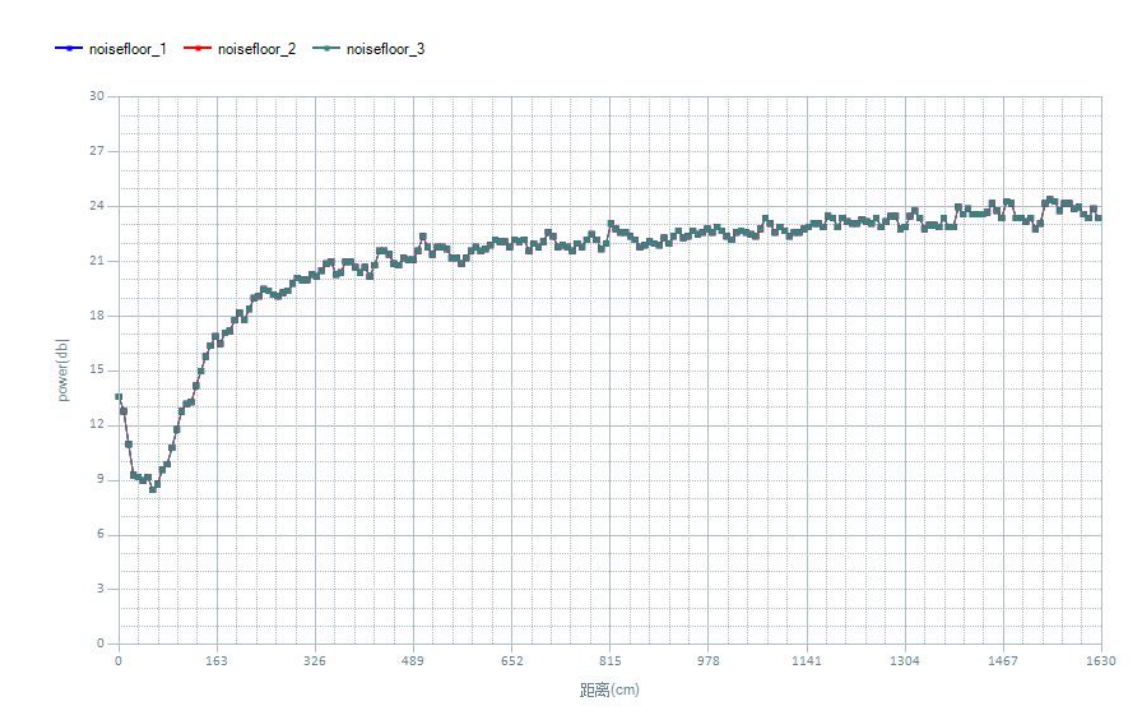


Figure 9-7 Normal background noise pattern

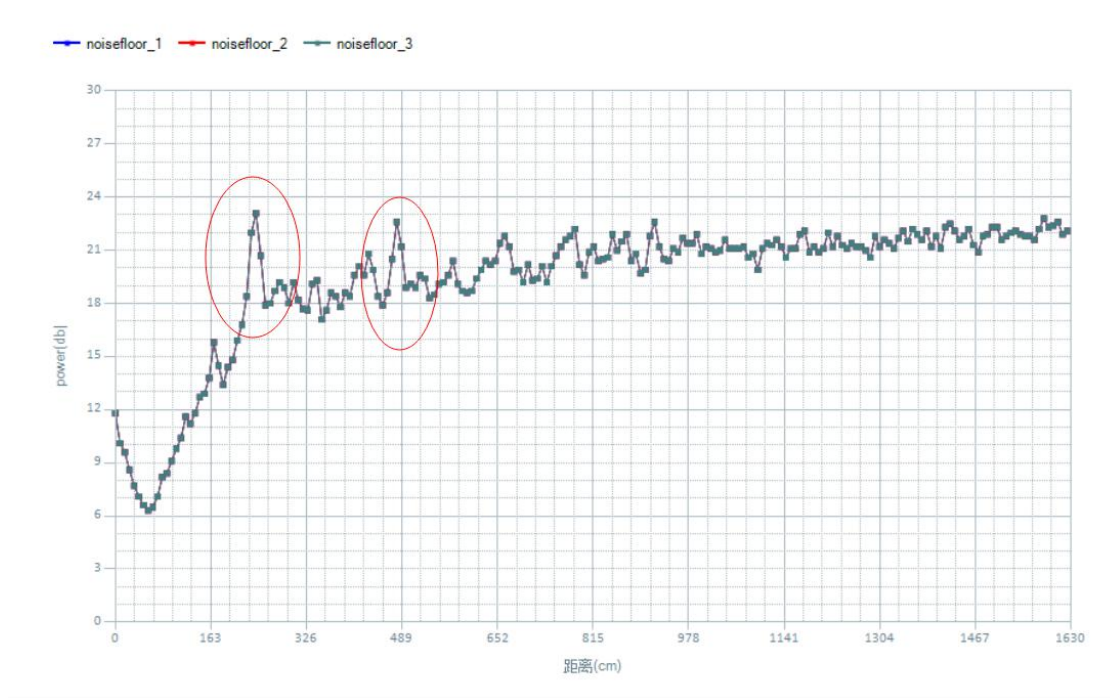


Figure 9-8 Abnormal background noise pattern

9.2 DCDC Output Performance Measurement

If the user's solution utilizes a DCDC, its output performance needs to be measured to ensure it meets the radar V13 power input requirements.

- When the radar is operating, it periodically emits millimeter waves to detect the environment. Therefore, the V13 power supply undergoes periodic power extraction. At the moment of emission initiation, a short-term voltage drop inevitably occurs in V13. However, the external circuit design should minimize the DCR (Distributed Capacitance Resistance) on the path from the DCDC output to the chip's V13 pin. It is recommended to ensure that the voltage drop is less than 50mV, with a maximum of no more than 100mV, and that the voltage after the drop remains above 1.2V.
- The switching noise of DCDC can couple to the V13 network, forming power supply ripple (excluding the voltage drop during the aforementioned emission start-up transient). Excessive ripple can interfere with radar RF. Therefore, during the design of external circuits, appropriate capacitors should be placed in the V13 power supply network to ensure that the power supply ripple V_{pp} measured by V13 is less than 50mV.

9.3 PCB Performance Verification Scheme

Table 9-1 PCB Performance Verification Scheme

Project	Validation Criteria
Power consumption	Using the actual measurement data from the development board under the same configuration as the comparison standard, the main focus is on the base current and average current
Various power supply quality	The drop amplitude of VCC is less than 100mV
	The drop amplitude of V13 is less than 50mV, with a maximum of no more

	than 100mV, and the voltage after dropping is greater than 1.2V
	The power supply ripple Vpp of V13 is less than 50mV
Noise floor	The extreme amplitude difference of different polylines at the same distance is less than 10dB
	The amplitude is below -72dB within 0~2m, below -76dB within 2~4m, and below -80dB within 4~10m
Performance from all angles	Program the DEMO firmware, perform distance measurement/angle measurement, and compare with the development board

9.4 Q & A

Table 9-2 Q & A

Project	Problem	Solution
Power quality of various sources	VCC drop is too large	Check the power supply capacity of the circuit
	V13 ripple is large	Check whether the output capacitance of V13 is sufficient
	V13 has a large transient drop	According to the BOM table in the PCB manual, check whether the DCR of the inductor/magnetic bead is too large, and whether the output capacitor is sufficient
noise floor	The background noise is too loud	Remove the DCDC and use internal LDO for power supply to confirm whether the issue is caused by the DCDC.
Performance from all angles	If the performance does not meet the requirements	Recheck power quality and background noise
		Check whether the PCB floor design meets the requirements

10 Design Checklist

Users are required to conduct self-inspection against the following check items before and after designing the schematic diagram/PCB.

10.1 Schematic Design Checklist

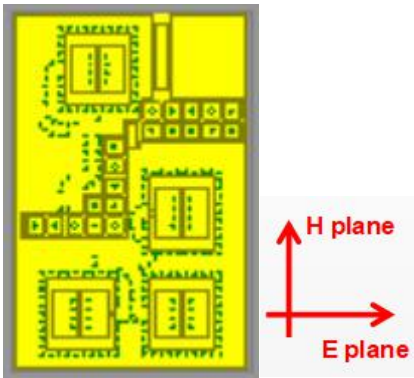
Table 10-1 Schematic Design Checklist

External connection signal	Communication interface	Supports UART/I2C/SPI communication. After selecting the interface, please refer to Section 7.1 for precautions and checks. If UART/SPI is chosen, it is necessary to ensure that the corresponding signals on the main control terminal (pin RX (PB1) for UART0, and pins CS (PA0) and CLK (PA1) for SPI0) are configured as pull-up or pull-down when the main control is in hibernation. Otherwise, the radar may experience abnormal leakage. If the main control cannot provide the pull-up/pull-down configuration during hibernation, it is necessary to configure pull-up/pull-down resistors on the periphery.
	Control signal	Regardless of which communication interface is chosen, it is recommended that the master control terminal allocate IO to control the RESET and PBO signals of the radar. When it is necessary to update the firmware of the radar online, the radar chip can be reset via the RESET signal. If PBO is detected as being pulled low during the reset process, the radar will enter firmware programming mode, allowing the main controller to update the radar firmware.
	Trigger signal	PA6/PA11 is preferably used as the signal for triggering the host.
	Test point	Reserve PBO, PB1, and PA0 for radar performance analysis. Our company provides radar analysis firmware and host computer tools. The radar needs to interact with the host computer through UART1-TX (PA0)/UART0-TX (PBO)/UART0-RX (PB1).
Power Design	VCC	Supports 1.7~5.5V, with 3.3V recommended. When the radar is operating, it periodically emits millimeter waves to detect targets, resulting in periodic fluctuations on the VCC. After adding a filter capacitor, the fluctuation is generally within 100mV. If there are voltage-sensitive devices on the periphery, such as a speaker amplifier, do not share the VCC with the radar.
	VDDIO	Supports 1.7~3.6V, with 3.3V recommended. If VCC is not greater than 3.3V, VDDIO can be connected to VCC.
		The flash is powered by VDDIO. If a VDDIO voltage below 3.3V is chosen, it is important to select a wide-voltage or low-voltage type of flash.

	V13	V13 can be powered by an internal LDO. If low power consumption is required, it should be switched to an external DCDC for power supply. The radar RF is sensitive to external DCDC switching noise. Please refer to the circuit module description in Section 7.2 to check whether the component selection is correct.		
	Other designs	The remaining power supplies can be designed based on the schematic diagram referred to in Section 7.2 . Please ensure that the capacitance values of the peripheral capacitors are consistent with those shown in the reference schematic diagram.		
Flash selection	Model Selection	Refer to the "Key Material Support List" to select the flash model. If a model not listed in the support list is required, you need to add the parameters yourself and conduct verification testing.		
	Capacity selection	If the internal OTA update function of the radar is not used "Internal OTA" refers to the process of first downloading firmware to the backup area stored in the radar, and then allowing the radar to complete the firmware update on its own.	Development-free solution	Choose a flash memory with a capacity of no less than 512KB
			SDK solution, i.e. customizing radar firmware	If the XIP scheme is not used, the flash capacity should be chosen to be no less than 512KB
				If the XIP scheme is used, the flash capacity should be larger than the size of the compiled firmware
			If the internal OTA update function of the radar is used	Then double the capacity for the scenarios listed under "Do not use OTA" and provide redundant space to store the new firmware
	Attention	If using WINBOND's flash, it is necessary to reserve pull-up resistors for MOSI/MISO to reduce sleep leakage at the flash end.		
Other	Refer to the device selection on the schematic diagram. Please refer to the instructions in Section 7.2 "Reference Schematic Diagram" to check whether the selection is correct.			
	The capacitance around the chip must be designed strictly in accordance with the capacitance values specified in Section 7.2 of the reference schematic diagram .			

10.2 PCB Layout Checklist

Table 10-2 PCB Layout Checklist

Floor layout	The GND layout around the radar chip significantly affects antenna performance. Please refer to Section 8.1 on PCB floor performance optimization for design guidance. Especially for solutions with cross-cutting needs or high angle measurement requirements, it is necessary to strictly adhere to the guidelines provided	
	The PCB structure surrounding the chip should be as symmetrical as possible, especially for solutions requiring angle measurement or cross-cutting. An asymmetrical structure will produce an asymmetrical directional pattern.	
Chip placement method	In the top-loading scenario, there are no specific requirements for chip placement.	
	In side-mounted scenarios, it is preferred to place the chip vertically, with the E-plane of the antenna in a horizontal direction and the TX antenna facing upwards. This configuration enables superior wide-angle detection capabilities. 	
Device layout	No DCDC solution	Refer to Section 8.9 LAYOUT Example - P1806 Module Layout for guidance. For single-sided layout, components are distributed on both sides of the short edge of the chip, with no components placed on both sides of the long edge.
	There is a DCDC solution	The radar RF circuit is sensitive to the switching noise of DCDC, so the external DCDC on the V13 power supply or other DCDCs in the solution should be kept away from the radar chip.
		The preferred layout is double-sided, and can be referred to in Section 8.10 LAYOUT Example - P1812 Module Layout .
	The capacitors around the chip are placed close to the corresponding pins, with a spacing of 2mm.	
Lamination	At least four layers of board	Top layer: Place the radar chip, and prioritize placing other components on the bottom layer. Avoid routing around the radar chip, and a complete GND is more conducive to improving angle measurement/cross-section performance.

		Layer 2: Set as a complete GND. It is not recommended to route wires on this layer.
		Layer 3: Routing as needed.
		Bottom layer: Place the remaining components.
Routing	VCC/V13 will have a peak current of 100~200mA, and the preferred trace width is greater than 0.5mm, at least greater than 0.3mm.	
	V13 and V18 are power-sensitive, and their routing should not overlap with signal lines and VDDIO on adjacent layers.	
	If SPI is used as the communication interface, the external SPI Layout needs to be routed according to Section 8.8 .	
	The routing width for the VCC input of the DCDC needs to be greater than 0.5mm, and the filtering capacitor for VCC should be situated close to the VCC pin of the DCDC.	

11 Product Packaging Instructions

11.1 Packaging Dimensions

*All dimensions are marked in millimeters

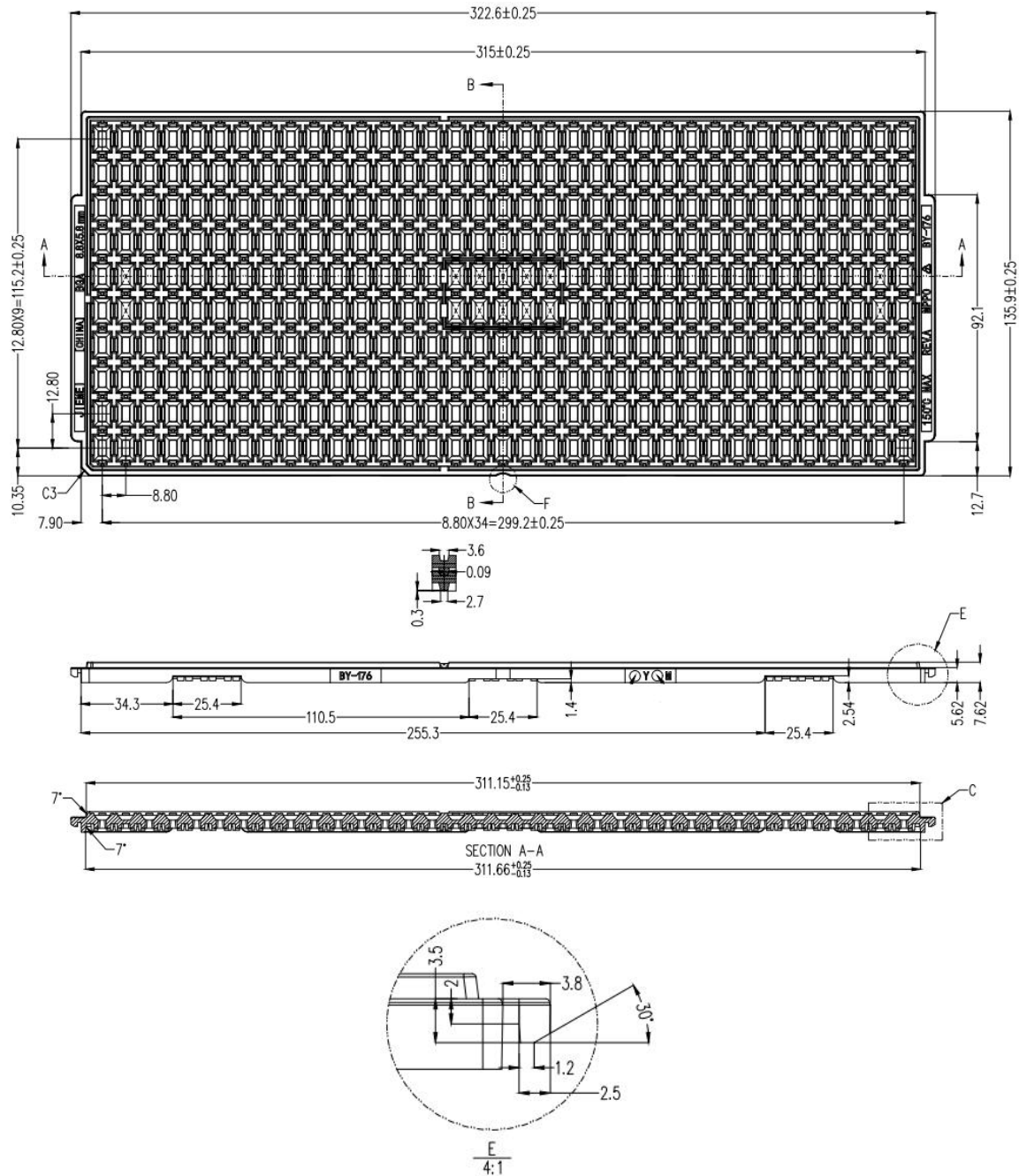


Figure 11-1 Schematic diagram of tray size I

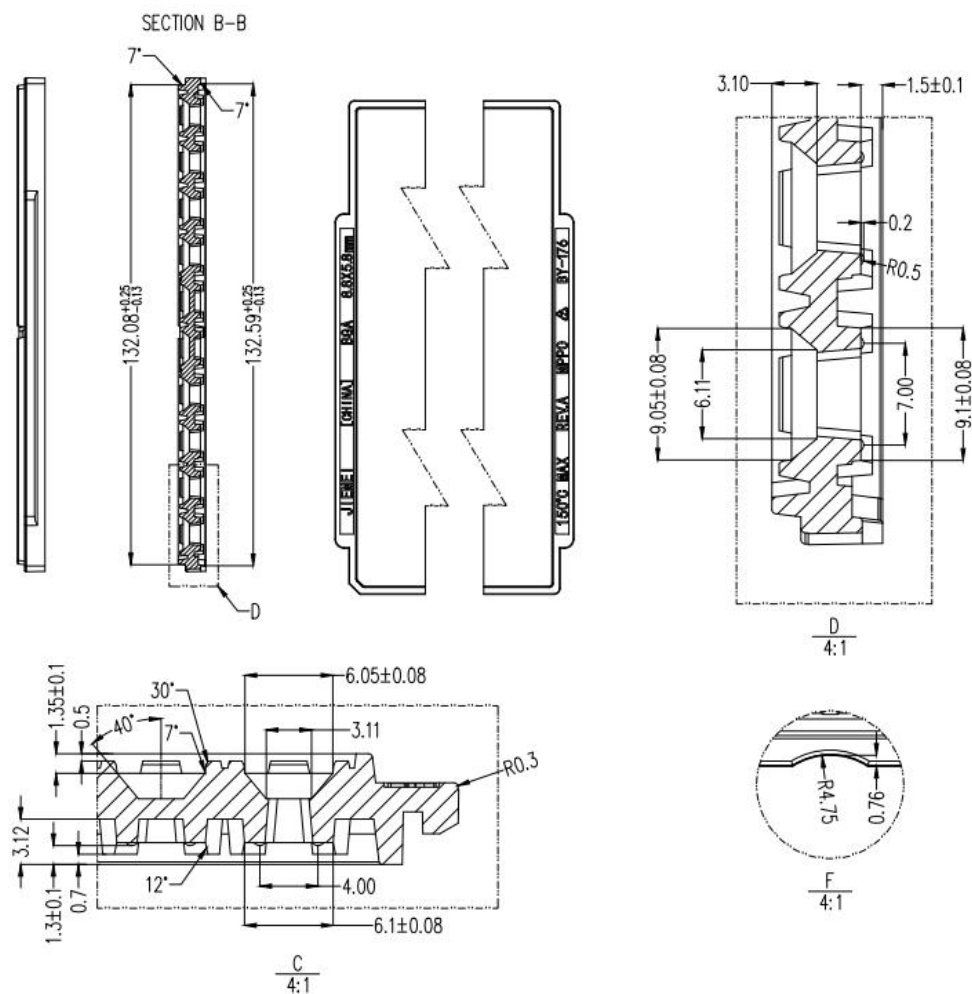


Figure 11-2 Schematic Diagram of Tray Dimensions II

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